Who Changes in College? Selection, Field of Study, and the Political Returns to Higher Education

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Abstract

Does higher education causally affect political behavior, or do the political choices of graduates simply reflect who was always going to enroll? This paper reframes the question as one of political returns, distinguishing engagement from partisan direction and asking how those returns are distributed, rather than trying to attain a perfectly exogenous measure of average effects. Extending the study of causal effects by comparing higher-education effects in the contrasting party systems of Germany and the United Kingdom. I ask how returns are distributed across predictors of university graduation, rather than abstracting from treatment selection as a theoretical nuisance to control for. Attainment shifts graduates toward the socially liberal and fiscally conservative parties, rather than the redistributive social democrats. However, this return is unevenly distributed, concentrating among the graduates most likely to attend in Germany and the least likely in the United Kingdom, dictated by electoral rules rather than educational differences. The credential alone produces similar shifts in every graduate; where they differ, it is due to their curriculum. The partisan returns belong to communicative-cultural fields, and the field of study governs whether their preferred party is taken to the ballot rather than whether it is formed.

Keywords: Higher education; socialisation; comparative politics; selection; heterogeneity; political engagement; party identification; left party support; field of study; difference-in-differences; Germany; United Kingdom

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Higher education shapes political behaviour along two dimensions: graduates' political engagement and partisan support. The literature has become defined by the asymmetry between the two. Causal designs accounting for pre-adult socialization find selection accounts for the majority of the educational cleavage in civic and political engagement outcomes: political interest, turnout, participation, and civic activity (22; 29; 24; 48; 52). Yet when it comes to political *direction*, that is, graduates' political attitudes, orientations, and partisan support, the effects remain robust, substantively large, and often traceable to the field the graduates studied. Field-of-study effects on direction are documented in both Germany (20) and Britain (36), with field-based decompositions extending the finding to immigration attitudes across European democracies (13). Authors disagree about whether field effects operate on selection, content, or context (39; 21). The disagreement itself runs partly along national lines. This complicates the traditional literature on educational cleavages (19; 48): once selection is accounted for cleavage in political participation between graduates and non-graduates is small or absent. Political impacts on an individual seem to lie more among the graduates themselves in which side they are on and what they studied to find themselves there.

This asymmetry calls for a change in what I refer to as the *political returns* of higher education. The dominant frame, descending from Putnam (30) and developed in the resource models of Verba et al. (47); Brady et al. (6), treats higher education as an investment in civic skills that improve one's capacity to articulate, participate and engage politically. *Political returns* brings into view what civic returns leaves invisible: political interest as attitudinal disposition rather than behavioural input (41; 31), party identification as attachment to a political community (52), and crucially, which party graduates support. Education seems to sort graduates towards supporting green and left-libertarians (1) with field of study explaining substantial variation in their immigration attitudes underlying these partisan choices (13). This points to a double sorting: politically engaged citizens sort into higher education, and higher education sorts them into different political camps depending on what they studied.

Changes among graduates by field face the same endogeneity concerns that attainment does, after all, intuitively, a field is chosen in the same breath as choosing to attend university[2]. In causal research on attainment effects both liberal attitudes (37) and engagement largely dissolve to selection. Engagement differences appear already in adolescence prior to university attendance in Germany, the Netherlands and the United Kingdom (52; 43; 38). However, field of study variation is detectable under fixed effects (36; 21). Yet this recent work has not directly compare the contributions of attainment and field of study to the same variables across different educational contexts. This paper asks the qeustion, between completing a university degree and studying liberal arts (or cultural and communicative content (44)), which contributes more to improving political engagement and partisan support? Any university credential or only certain fields of study?

If education is a proxy for prior political dispositions, the proxy works on the socio-cultural dimension and not on the economic dimension (37; 38). The educational cleavage has since the expansion of higher education been increasingly associated with new left-wing parties or issue-owning parties such as the Greens in Germany (1), rather than the old social democratic parties. These established associations are instructive because if there is a causal impact of tertiary education, this literature would expect the graduates to shift towards newer, issue-owning parties such as the green left-wing rather than redistributive social democratic parties. This leads to the prediction that, tertiary attainment leads to support for green left wing parties rather than support for left-wing parties generally.

This paper examines two theoretically distinct but inseparable educational choices as treatments: choosing to get a degree, and which field (operationalized as communicative and cultural fields versus economic and technical fields (CECT) for a detailed review, see Van de Werfhorst and Kraaykamp (44); Hooghe et al. (20)). Each one involves endogenous choice, moreover, the choices are predicted by mostly similar socio-demographic characteristics, but they operate on political outcomes through different mechanisms. This paper approaches these two choices using two datasets that complement each other well: The German SOEP and the British UKHLS. In the SOEP, field of study is observed as the actual completed field, capturing the effect of curricular

[2]I understand that this doesn't always apply to all national contexts, where you can decide your major a year after enrolling in university, however courses are also selected.

exposure; in the UK, field is observed in a pre-enrolment preference stated at age sixteen, capturing a weak intention-to-treat effect. The two countries also differ in their educational and electoral regimes. They share a requirement for academic secondary school qualifications, Abitur and A-levels (or International Baccalaureate). These two different country contexts provide fertile ground for investigating how mechanism of tertiary and field effects can vary across political contexts.

Existing Research

The causal effect of higher education and fields of study on political behaviour and orientations remains intensely debated due to mixed empirical findings and the endogeneity issues that comes with estimating causal effects of educational choice (25; 39; 48). Identification strategies that seek to exploit "as if random" variation, such as regression discontinuity designs based on compulsory schooling laws (11), have yielded mixed results; while some detect small causal impacts of fields of study on political orientation (21), attitudes (14; 39; 36), or behaviour (2). The contested nature of these effects is well illustrated by the exchange between Kam and Palmer (22), who used a matching technique to argue that political participation was a spurious product of selection, and Henderson and Chatfield (17), who demonstrated that this result was driven by poor covariate balance and a handful of outlier matches. The underlying hurdle is the same in each case: attainment, attendance, and choice of field of study are all endogenous, and all can act as proxies for latent variables such as gender (12; 46), pre-adult socialization, and family background (29; 25; 37; 35).

Bourdieu (5) argues that higher education functions as a legitimising force of existing class hierarchies by sorting individuals into the social positions they are presupposed to occupy. Evidence of this thesis is found in the sorting model of education, or education-as-proxy argument (29; 48), wherein the only change after controlling for pre-adult factors reflects the existing social hierarchy, a consequence of social sorting rather than an individual mechanism of socialization. According to Campbell (10) and Persson (28), the observed correlation between political engagement and tertiary attainment reflects the underlying characteristics of graduates more than it reflects a genuine causal impact. This is a compelling argument, but it is not a sufficient reason to conclude that socialization plays no role. Rather, it suggests that I cannot properly evaluate socialization without first modelling the heterogeneities of selection into higher education and how these vary within higher education across the fields of study people chose — treating these choices as integral to the causal process itself. Given the frequent finding that most liberalization processes are products of self-selection (37), it is worth investigating whether this self-selection is uniformly distributed across the likelihood of graduating higher education and across fields of study.

This brings us to a core theoretical problem. If I want to understand socialization, what higher education actually does to a person, I must first ask who is being socialized. The question of effect is inseparable from the question of selection, and selection into tertiary education is neither random nor uniform. The heterogeneity of who enters university, and under what conditions, shapes the distribution of any socialization effect I might hope to observe. Yet it remains an open empirical question whether selection patterns are themselves consistent with the socialization hypothesis (negative selection, whereby those least likely to graduate change the most) or the selection-as-proxy hypothesis (positive selection, those more likely to graduate change the most upon doing so because attainment amplifies pre-existing views or trends in engagement), or some combination of the two. Difference-in-difference models are limited in their capacity to investigate such patterns. This heterogeneity requires mapping a likelihood of graduation, and then using those stratifications to discover whether effects are uniform or concentrated among "particular types" of people as Simon (38) indicated.

As a recent review points out (49) the endogenous decisions in a causal process are integral to understanding the outcome, and requires investigation. The heterogeneous treatment effects of education on political behavior may stem not only from the educational experience itself but also from the endogenous selection into both tertiary education *and* field of study, both choices significantly impacted by prior characteristics. The recent focus of political scientists engaging with the black box of endogeneity that is *causal effects of higher education* (37; 36) or *causal effects between fields of study* (25; 20) often leads to a pre-occupation with isolating, accounting for, abstracting from or establishing the relative contribution of self-selection. Selection in this context is understood in terms of the latent effect of individual background characteristics that can predict both political behaviour and these two educational choices. Yet, surprisingly, few have focused on variations in these political changes in terms of their background characteristics. In other words, I do not know what this selection is driven by; it is the estimated remainder, which I attribute to individual priors. Studying, for instance, the political change of randomly assigning students to study humanities is going to produce a different result than if they themselves had chosen to participate in such a class. Thus, I propose to focus on the student's decisions as part of the causal process, and using a prediction of their graduation to understand who is likely to be causally impacted by higher education, and who is likely to be impacted by their choice of discipline.

Tertiary Attainment and the Education–Politics Link

Brand (7) demonstrates that university education has non-uniform effects on the civic improvement I can expect from college completion. Specifically, improvements in civic engagement resulting from college completion decrease as the predicted likelihood of completing college increases. This pattern is consistent with what Brand and Xie (8) refers to as negative selection, that is, where individuals who are least likely to complete college but nonetheless do are most likely to improve. In simple terms, people who were unlikely to attend and graduate from university, but did anyway, gain the most from doing so. The comparison group consists exclusively of high school completers, as dropouts are excluded from the sample. The crucial aspect here is establishing a rich set of pre-treatment characteristics in order to define low and high propensities for completing university. Low propensity, for instance, was defined by parents with limited formal education, lower family income during adolescence, a non-intact family structure, having four or more siblings, being Black or Hispanic, lower scores on prior cognitive indicators, enrolment in non-college-preparatory secondary school tracks, and having peers with limited educational plans or parents who did not encourage college enrolment.

The conclusions of Brand (7) support the Verba et al. (47) resource explanation in terms of the two resources that also predict skill acquisition (recall the three resources theorized to improve civic engagement by Verba et al. 1995: time, money, and skills). Those with a high propensity for college attendance also have more of these three resources, as well as an inherited politically engaged network. Their adolescent environments cultivate political interests earlier and more strongly than those of their low-propensity counterparts. A first-generation working-class tertiary graduate gains all three resources and a network, while a wealthy graduate student from a family of doctors is unlikely to experience the same change. Using this example, I can see how the differential access to resources and networks prior to tertiary attainment informs both the *decision* (selection) and the impact (causal). This differential has a further implication for the present moment. The share of parents who have attained tertiary degrees among students enrolling in higher education has naturally also expanded; one would therefore expect that in terms of within-individual change, the acquisition of civic skills has become *more* important than when the resource model was first theorized (47), precisely because the inherited resources that once did much of the work are less unevenly distributed across the student population than before.

Tracked Secondary Education and Field of Study

Before reviewing the literature on tracked education, two distinct concepts that structure the analysis should be clarified. Secondary school *track* refers to the institutional pathway through which students are sorted during compulsory education — Gymnasium versus Realschule/Hauptschule in Germany, A-levels versus vocational qualifications in the UK. Track determines who is eligible for university entry and is therefore central to the propensity score for tertiary attainment. *Field of study* refers to the disciplinary content of the university degree itself, classified here using the CECT framework (Communicative, Economic, Cultural, Technical) of Van de Werfhorst and Kraaykamp (44); the operational definition — the communicative-cultural skill ratio and its dichotomisation — is set out in the research design. The distinction between track (which determines *whether* someone enters university) and field (which determines *what* they study once there) is analytically central: the paper estimates separate propensity scores and treatment effects for each decision.

With the expansion of tertiary education, I also identify new potential sources of heterogeneity, such as the theory that specific competencies within fields of study can socialize particular sets of attitudes or behaviors (44). Recent results by Hooghe et al. (21) demonstrate these field effects when examining Dutch and German graduates who majored in fields with more communicative or cultural competencies. However, while they identify socialization effects on GAL party support, their effect sizes are substantively minuscule. These effect sizes could be due to the fact that, in their estimation of attainment as an event, they average a dynamic treatment effect that is concentrated in a particular period across enrolment cohorts and periods. Contrasting graduates with non-attainers within a track-field strata is a compound treatment. On one hand, the credential, the university experience in a certain period, and university-level curricular content arrive together. On the other hand, a control whose own vocational or non-attaining trajectory reflects either inaccess or a choice of not proceeding, both of which are consequential to their politics. That design does not attribute its estimate to curricular content as such. Isolating content means holding attainment and the university experience fixed, comparing graduates of one field with graduates of another, and the counterfactual of the same graduate having studied something else.

Research on tracked education systems reveals outcome-dependent findings regarding political socialization by academically or vocationally oriented tracks. When examining immigration attitudes among Swiss youth from age 13 to 30, Lancee and Sarrasin (23) found no indication of tracked socialization patterns. Comparative studies (50) find some evidence linking secondary school track effects to higher education: tracked systems reduce enrolment in higher education, which reduces the pool of civic skills and political knowledge and thus civic and political engagement. Even among adolescents, strong selection patterns in political interest appear early. Zhang et al. (52) examined difference-in-differences of students by their track in the German National Educational Panel Study, showing that students had different levels of political interest *prior* to enrolment in upper-secondary education, with no significant causal effect of academic tracking on political interest thereafter. This is consistent with new data from the Dutch Adolescent Panel on Democratic Values, where Van De Werfhorst et al. (43) find that although students increase their political knowledge and trust in institutions in more advanced tracks, track differences in civic and democratic attitudes are already present before secondary education begins.

While the British comprehensive education system is not tracked to the degree that the German system is, it is however differentiated through the A-levels, which build on the generalized schooling of GSCEs. Similar selection patterns are found in research on the British higher education system, although with consistently modest causal effects leftward on cultural attitudes (25; 38; 35): moreover, students with culturally liberal priors disproportionately enroll in humanities and social sciences (HSS), while those with right-leaning economic views often select into STEM or business fields (39; 36). This university effect is further moderated by the system's high degree of mobility, since the vast majority of full-time students live in either university towns or more cosmopolitan areas, a similar environment conducive to peer socialization, moving towards the population political mean of their new neighbourhoods (40). Similarly to the German case, the long-term political divide between graduates and non-graduates in Britain is attributed to several factors; selection, socialization, curriculum, the allocation into new social networks and high-skilled workplaces that for example reinforce a less Eurosceptic and more culturally liberal worldview (25).

Hypotheses

The paper sets forth four predictions. Firstly, if educational attainment reflects only pre-adult dispositions, the treatment effect of tertiary attainment should collapse if I condition on pre-treatment political outcomes. Secondly, supposing there is a causal leftward shift from university graduation, I should observe support concentrating among socially liberal green and left-libertarian parties rather than a generalized left-wing support (1; Hix). If the findings by Brand (7) apply, treatment effects will mainly help graduates from disadvantaged backgrounds. However, if we see that more advantaged students benefit instead, it suggests a mechanism of cultural amplification rather than substitution. Finally, if the curricular content of communicative and cultural fields drives the partisan effect, the contrast between high-CECT and low-CECT graduates produces a partisan effect distinguishable from the attainment effect alone.

Table 1: Hypotheses

	Hypothesis
H1	Tertiary attainment has a causal effect on political behaviour, independent of pre-adult selection.
H2	University graduation shifts partisan support toward green and left-libertarian parties, not toward the left as a whole.
H3	The political return to attainment varies systematically with the likelihood of attaining a degree.
H4	A field's communicative-cultural content shifts partisan behaviour independently of attainment.

Research Design and Data

Rather than abstracting from the endogeneity of educational choice, this paper tries to treat the endogenous choice as a meaningful inference. In tracked educational systems like Germany, tertiary attainment is strongly predicted by family background, gender, and migrant status. Whereas in the United Kingdom, such predictions are weaker. If education's political consequences vary by the likelihood of treatment, that variation tells us something about how the institutional structure of accessibility determines socialization into politics. With that in mind, the research design asks how returns to higher education vary across predicted access to it. And how do these returns vary between different institutional configurations of the same selection process?

What would the average treatment effect be under random assignment? Is not a question this design seeks to answer, nor is it one that can actually extend to the general population. Education is a service that is restricted and predicted by several factors; this design treats these factors as substantively interesting sources of variation rather than something our estimator has to be robust to. Higher educational causal effects on political participation tend to converge towards a null finding (28; 24; 37). This is partly because of how aggregate ATTs are defined, we cannot recover effects from people that would have quite different counterfactuals. Someone that is unlikely to attend university relative to someone who is almost certain to attend university experience and choose the same thing, tertiary education. Lindgren et al. (24) demonstrate that improving access to higher education increases turnout for lower SES individuals and has no effect on higher SES individuals. Brand and Xie (8) and (7) frame the same intuition for higher education. Our design treats this source of heterogeneity as an estimand rather than a nuisance parameter.

Data

The analysis draws on two nationally representative household panels: the German Socio-Economic Panel (SOEP, 1984–2023) and the combined British Household Panel Survey and UK Household Longitudinal Study (BHPS–UKHLS, 1991–2022). Germany and the United Kingdom represent qualitatively distinct higher-education regimes, this contrast is central to the research design. Both panels include youth questionnaires that provide pre-treatment political and educational measurements during adolescence. The data was restricted to cohorts born in or after 1975 in order to restrict analysis to the post Bologna 1998 reform and the large higher educational expansion of 1998 up to today. This restriction yields $6,988$ treated and $28{,}211$ never-treated in Germany and $1,741$ treated and $3{,}614$ never-treated in the United Kingdom (for a full breakdown see Table 2).

Treatments

Treatments are binary, absorbing, within-person, time-varying indicators that switch to 1 from the year of first observed higher-education enrolment and stay treated from there on. This operationalisation is harmonised in tertiary attainment across the two panels. Field of study however is only possible to measure in the SOEP[3]. Field treatment follows the definition of CECT classification (Communicative, Economic, Cultural, and Technical) (45; 44), and the dichotomised procedure of these scores by Hooghe et al. (20), at the within-sample median of the CECT ratio. The CECT ratio is defined as the ratio of communicative and cultural skill content relative to the sum of all four

[3]Field of study is recorded in the SOEP after a degree is obtained, it is carried forward and backward within individuals in order to achieve a corollary measure to tertiary treatment.

Table 2: Analytical samples by specification

Specification	n_{treated}	n_{control}	Comparison
A. Tertiary attainment, pooled			
SOEP	17,583	38,560	HE enrolment/completion vs no HE
UKHLS	9,666	29,320	HE-equivalent qualification vs no HE
B. Tertiary attainment by field track (SOEP)			
Tertiary | High-CECT track	1,374	3,054	HE effect within communicative-cultural
Tertiary | Low-CECT track	1,872	17,478	HE effect within economic-technical
C. Field of study within tertiary completers (SOEP)			
CECT-DE: high vs low CECT	1,322	2,752	Communicative-cultural vs economic-technical
D. Robustness specifications			
SOEP — Tertiary (Abitur/Fachabitur)	5,647	3,790	Academic-track sample restriction
UKHLS — Tertiary (A-level only)	5,901	2,084	Academic-track sample restriction
SOEP — Tertiary (voc. controls)	9,269	7,399	HE vs vocational-training completers
SOEP — Tertiary (age $\leq$ 31)	6,878	24,292	Age restriction
SOEP — Tertiary (born $\geq$ 1980)	6,988	28,211	Narrower-cohort robustness

Notes: Unique persons in each specification; outcome-specific cells in partisan-direction analyses are smaller because they condition on reporting party support.

skill dimensions (communicative, economic, cultural, technical). With values taken from Van de Werfhost and De Graaf (45)'s graduate survey; full formula in Appendix .

The control group is the never treated population, in Germany this encompasses both those who obtained an Abitur or Fachabitur but did not proceed to university, and those who completed vocational training (Ausbildung) without subsequent tertiary enrolment, and those whose highest qualification is upper-secondary or below. In the United Kingdom, the equivalent group comprises A-level holders who did not proceed to higher education or equivalent qualifications, e.g., ISCED-4 or below. Any respondent observed at any wave to hold a higher-education qualification at Bachelor level or above is in the treated group. The main comparison is between tertiary attainers against non-tertiary attainers. Two robustness tests are reported to this. The first restricts the control group to only vocational groups, in order to test difference between post-secondary education. The second restricts the entire sample to the academically-prepared subpopulations: Abitur or Fachabitur holders in the SOEP, and A-level holders in the UKHLS. This tests whether the main effect survives among those most likely to be treated based on their academic preparation.

For the field of study treatment the paper reports two comparisons (Table 2, Panels B and C). Panel C is the estimand of interest: it narrows the sample to tertiary graduates and contrasts high-CECT against low-CECT graduates. Because both groups hold a degree, the comparison isolates the effect of communicative-cultural content within the university, so that the counterfactual is the same graduate had they completed a low-CECT degree. Panel B instead estimates the tertiary effect separately within the high-CECT and low-CECT tracks against non-attainers, following Hooghe et al. (20): the low-CECT never-treated group is dominated by those whose highest qualification is Hauptschule or below and by vocational trainees in technical and engineering occupations, the high-CECT group by trainees in medical, caring, teaching, social, and artistic occupations. In Panel B the attainment effect is estimated and reviewed on whether or not it varies by track.

Outcomes

I examine political engagement and behaviour as two different dimensions of political returns to higher education and field of study, harmonised all covariates across the SOEP and UKHLS using the manual developed by Turek and Kalmijn (42).

The first dimension concerns two engagement items, political interest and party identification. *Political interest* is measured annually on a four-point scale in both countries (1 = not at all interested, 4 = very interested). *Party identification* is a binary question, asked separately from which party they identify with, the respondent indicates whether or not they have a party they support.

The second dimension relates to the direction of political behaviour. I use the political party support variables in combination with retrospective voting record and party identification variables in the UKHLS and SOEP to distinguish leaning towards a party from voting for a party. This illustrates the slight discrepancy between the waves in terms of which party panel respondents recall they voted for and what party they supported in the year the election was called. Concordance between lagged support variable and recall vote choice ranges from 0.89 to 0.95.

In order to investigate party support in terms of respondents' realised political behaviour: I constructed an annual binary composite for each party. In the SOEP the composite has three tiers:

a recall vote item from the waves following three general elections in Germany, waves 2014, 2018, and 2022 (`plh0333`: "Which party did you vote for in the 2013 / 2017 / 2021 Bundestagswahl?"); a party support item given they answered yes to the engagement question about party identification at all other annual waves (`plh0011`: "Generally speaking, do you lean towards a particular party?"; followed by `plh0012`: "Which party do you support?"); and zero for respondents reporting no party identification. In the SOEP respondents were asked to indicate which party they voted for, in the UKHLS respondents were asked if they voted in the last general election (`vote9`: "Did you vote in the most recent general election?"). The UKHLS composite variable is constructed from a combination of party support (`vote1`: "Generally speaking, do you think of yourself as a supporter of any one political party?"; followed by `vote4`: "If there were a general election tomorrow, which political party do you think you would be most likely to support?") plus commitment proxy where respondents have reported which party they support prior to a wave where they participated in an election. Respondents in both datasets are therefore required to have voted in at least one general election.

Identification strategy

The design follows the standard approach of estimating aggregate treatment effects before investigating heterogeneities. Formal model specifications and identifying assumptions for each estimator are stated in the appendix ().

a) Aggregate effects of tertiary attainment on engagement and party support

Firstly, the aggregate effect of tertiary attainment on engagement and direction. I estimate the effect of university on engagement and party support using the staggered difference-in-differences **?**) estimator with inverse-probability weighting (IPW), never-treated controls, and bootstrapped standard errors. In this approach, people are under treatment from the year they first started their university education. The weights are constructed on predictors of higher education: gender, parental education, migrant background, and year of birth. Cohort-time treatment effects are then aggregated to an overall ATT of attending university (9).

For the United Kingdom, I estimate the same outcomes using the Xu et al. (51) interactive fixed effects (IFE) estimator on the per-wave absorbing status indicator, with covariates gender, ethnicity, parental education, and year of birth. UKHLS records the first wave, a qualification is reported rather than the year they enrolled and graduated. Fixed effect counterfactuals is a heterogeneity robust estimator that does not depend on cohort decomposition. Between the two different ways of estimating difference in differences, they have some different but testable assumptions: Parallel trends conditional on covariates and no anticipation of treatment effects in the enrolment-cohort decomposition for Germany, and the imputation of counterfactual parallel trends for the United Kingdom. I test both of them using pre-trend tests and a placebo equivalence test on minimum two periods prior to enrolment (32).

In addition to this aggregate analysis on party support, the composite measure (section) is estimated using IFE on both samples to test for discrepancies between the support and composite ATTs. This would identify whether university graduates' attachment to a party translates into voting for that same party. Additionally, the composite ATT is compared to the cohort-specific ATTs from the staggered specification to see whether any support-vote discrepancy is uniform across treatment cohorts or concentrated in particular cohorts, in order to examine cohort replacement variation in how attachment maps onto vote choice. Identification and assumption tests follow the same procedure outlined above for the UK data.

b) Heterogeneity by likelihood of treatment

Using the SOEP we can identify how the aggregate effect varies across the propensity to attain tertiary using the marginal treatment effect framework developed in Zhou and Xie (54). The framework distinguishes two dimensions of selection: Firstly, the observable propensity to attain a tertiary degree, predicted by demographics, family background, and the secondary school track. This is what Brand and Xie (8) uses to examine variation in treatment along likelihoods. Secondly, the unobserved resistance that is not captured by observables, people who defy their predicted treatment status for reasons not captured by the data. A respondent is treated when their propensity exceeds their resistance. Respondents whose propensity is well above their resistance are treated for predictable reasons; respondents whose resistance is well above their propensity are untreated for predictable reasons. Neither of these is particularly interesting for policy; the interesting population lies between them. People who are close in both their propensity and resistance sit at the margin of treatment. People in this group would be affected if the state made higher education more or less accessible.

Earlier approaches to estimating treatment effects by likelihood of treatment demonstrated an insufficiency (53). They were unable to distinguish whether the effect of a university degree varies by the likelihood of getting one because different people respond differently to treatment, or because we have not measured their drive or ability sufficiently. Due to the lack of biographical data on academic achievement, we cannot defensibly use the approach of Brand and Xie (8). The redefined marginal treatment effect (MTE) framework separates the two sources by using the local instrument.

The average treatment effects are estimated for three groups: people who received treatment, people who did not, and the marginal subpopulation. The last group, is the policy relevant treatment effect and yields an ATE for respondents at the margin of propensity. The local instrument used to predict treatment assignment is the birth region (Bundesland), which adds substantial information to the selection equation (section). The negative selection effect, those who are unlikely to get treated benefit the most from it, is stated in this framework as when the marginal effect declines with propensity, and vice-versa for positive selection.

The UK data does not have an equivalent instrumental variable (tuition-fee instrument failed the relevance test), instead the treatment effect by likelihood is approximated using a combination of data-driven approaches. Firstly, DiD ATTs within propensity quintiles are estimated, essentially this is a modern DiD implementation of the Brand (7) model. Secondly, the IFE estimator is refit within the strata of people that have their A-levels (i.e. the strongest predictor/the high propensity population). Finally, the IFE is estimated within cohorts by the year of treatment, early enrolment students tend to be high propensity relative to later enrolment students within a cohort. Should these three decompositions of the ATT converge, it would suggest non-parametric evidence of heterogeneity-by-likelihood patterns, applying a similar logic as the MTE (54).

A descriptive estimate complements this analysis, heterogeneities expressed in the aforementioned observable predictors used to calculate propensity scores. This identifies the mechanism described in Lindgren et al. (24) that electoral returns to tertiary attainment are largest among respondents who based on their family background are unlikely to attain a degree. I estimate the within ATT using augmented inverse-probability-weighted interaction estimators on each treatment-outcome-moderator separately. Moderators are gender, high parental education, migrant or ethnic background, and high parental occupational status. I report it as a descriptive complement to the estimates above.

c) Heterogeneity by field of study

The field treatment is a binary absorbing treatment indicator that considers people treated when their degree is above the median CECT ratio defined in Hooghe et al. (20). The within-graduate contrast (Panel C) defines those tertiary students below the median CECT as a control. In this panel the attainment effect, institutional exposure, and selection into university are all held fixed and the counterfactual is the same graduate had they completed a TE rather than a CC degree, isolating curricular content. Panel B, which follows the treatment definition of Hooghe et al. (20) contrasts attainers within a track, i.e. where any resulting ATTs reflect a compound comparison of content *and* attainment. Panel B answers whether the attainment effect varies by track, not whether content itself has an effect. Identification of the field effect rests on conditional ignorability of field choice given the observable propensity score.

The first step in estimating field effects is to stratify the SOEP by field track and estimate the tertiary attainment effect within each track (Panel B). The second step narrows the comparison between tracks within the university; this eliminates the tertiary attainment effect, and thus should be interpreted as a contrast between graduates by their exposure to communicative and cultural content (Panel C). The first step answers the question of whether the size of the tertiary attainment effect varies by track. The second step answers whether the type of degree matters among people who have completed one.

Each of these estimations uses the same CECT definition as Hooghe et al. (20), whose analysis on the same German Socio-Economic Panel runs the (IFE) estimator separately on high-CECT and low-CECT subsamples. The first step follows their design with a change in estimator, replacing IFE with the staggered difference-in-differences estimator of (a). The second step extends their panel analysis to a comparison they do not estimate: high-CECT graduates contrasted directly against low-CECT graduates. Their within-stratum identifying contrast is always tertiary attainment, never a direct high-CECT vs. low-CECT graduate comparison, i.e. this estimand does not appear in their design.

The estimator change is the cohort-decomposition argument of (a) applied to this layer. IFE averages a dynamic ATT across all cohorts at each event-time, regardless of what year that was, collapsing cross-cohort variation in the field effect into a single curve. The staggered DiD design identifies $\mathrm{ATT}(g,t)$ at each cohort separately by the year they enrolled, and aggregates to an overall ATT and cohort-summary ATT using explicit, non-negative weights over the cohort-time grid (15). This is particularly useful for understanding the qualitative change in educational expansion, for example, communicative-cultural graduates from the 2005 treatment cohort can have a different ATT than those from the 2018 cohort. The IFE cannot replicate the same cohort decomposition without refitting the estimator within each cohort. The two estimators otherwise agree on the headline within-individual ATT and both avoid the negative-weighting issues of two-way fixed effects (3; 33). The UK panel does not record completed field with sufficient coverage to support a parallel analysis.

Results

Aggregate effects on engagement and party support

Table 3 reports the ATT of tertiary attainment on engagement and party support in both panels, estimated as a doubly robust staggered DiD for Germany and using the counterfactual interactive-fixed-effects estimator for the UK. In Germany, graduating from university raises political interest by roughly four percentage points and party identification by six. The composition of party support shifts toward the centre-left and towards the Greens, with no equivalent movement towards die Linke or the CDU/CSU. The pattern reflects a shift towards issue owning parties with less support for economic redistribution than traditionally redistributive left-wing parties. In the UK the engagement effects are roughly twice as large as our German equivalent, harmonised on the same scales (42). In the UK, the partisan shift is concentrated towards the Liberal Democrats rather than towards the traditional social democratic party, Labour. Attainment in the tracked education system of Germany is overdetermined by prior characteristics such as family background, dem and social status **??**. Due to this characteristic the marginal effect on engagement is smaller than in the comprehensive UK system, where attainment is closer to a population-average treatment.

The composite vote-proxy ATT (Table 4) anchors party leaning in actual vote choice and clarifies a discrepancy in the UK partisan results. The composite voting variable for the Labour party substantially exceeds its support-leaning ATT. Graduates' Labour support lags behind their electoral performance, this is largely a reflection of strategic voting in a historically two-party first-past-the-post structure. The German pattern is more consistent between party support and the composite variable.

How effects vary by likelihood of attaining a degree

I identify the attainment effect on engagement and partisan support by likelihood of graduation, i.e. the political returns to higher education (in a similar vein as (8)). Two complementary identifications recover this heterogeneity: a local-IV implementation of the marginal treatment effect for

Table 3: Aggregate effects of tertiary attainment on engagement and party support

	Engagement		*Party support*			
	Political interest	Has party	Centre-right	Centre-left	Greens/LibDems	Far left
Panel A: Germany, Staggered DiD with IPW						
Tertiary	$+0.041^{**}$	$+0.061^{***}$	-0.032	$+0.056^{***}$	$+0.041^{**}$	-0.016
	(0.018)	(0.012)	(0.020)	(0.018)	(0.017)	(0.010)
Panel B: United Kingdom, interactive fixed-effects counterfactual						
Tertiary	$+0.087^{***}$	$+0.093^{***}$	-0.027	$+0.003$	$+0.031^{**}$	$+0.011$
	(0.022)	(0.013)	(0.017)	(0.021)	(0.015)	(0.010)

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$; bootstrap standard errors in parentheses. Outcome columns harmonise across countries: *Centre-right* is CDU/CSU (Germany) and Conservative (UK); *Centre-left* is SPD and Labour; *Greens/LibDems* is the German Greens and the UK Liberal Democrats; *Far left* is Linke (Germany) and the Green Party (UK). Panel A: Staggered DiD with IPW and a never-treated control; Panel B: Xu et al. (51) interactive fixed-effects counterfactual on the absorbing-status indicator. Covariates throughout: gender, parental education, migrant or ethnic background, year of birth. Sample definitions in Table 2.

Table 4: Vote-proxy composite: interactive fixed-effects ATT, both panels

	Centre-right	Centre-left	Greens/LibDems	Far left
Panel A: Germany				
Tertiary	$+0.001$	-0.015	$+0.036^{**}$	$+0.007$
	(0.014)	(0.014)	(0.017)	(0.009)
Panel B: United Kingdom				
Tertiary	$+0.006$	$+0.088^{***}$	$+0.010$	$+0.026^{***}$
	(0.008)	(0.012)	(0.006)	(0.006)

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$; bootstrap standard errors in parentheses. Interactive fixed-effects ATT on the annual vote-proxy composite, which combines lean between elections with recall vote at election waves (SOEP 2014/2018/2022; UKHLS 2010/2015/2017/2019/2024). Outcome columns as in Table 3.

Germany using birth Bundesland as instrument, and a three-way non-parametric decomposition for the UK in the absence of good instrument candidates in the data.

Table 5 reports the local-IV estimates for Germany following the Zhou and Xie (54) selection on unobservables criticism of the original heterogenous treatment estimator (8). The effect on political interest is essentially constant across the propensity distribution, improvements in engagement from attainment reflect treatment effects and not selection patterns. Graduates from backgrounds that with lower or higher likelihoods of graduating made the same gains. Partisan outcomes on the other hand demonstrate real change. The counterfactual effect on the untreated, that is, the partisan shift a non-attainer would have shown had they attained, is substantially larger than the effect on the treated, and the marginal effect rises across the propensity distribution. These patterns across propensities reflect positive selection patterns in support for the Greens in Germany. Graduates from more advantages backgrounds, likely to graduate from Univeristy, support the Greens more than than graduates from disadvantaged background. The pattern replicates the direction of the Brand–Xie propensity-by-treatment interaction on the same panel (Appendix Table 13), now identified under the local-IV assumption rather than under the assumption of zero omitted variable bias (selection on observables). These estimates are robust within a West-Germany sub sample robustness (Table 5, lower rows), the propensity gradient is not a legacy of East/West composition[4]. Full estimates for the four remaining outcomes are in Appendix Table 19, and the Bundesland fixed-effects DRDID robustness in Appendix Table 20.

For the UK the three converging non-parametric decompositions demonstrate similar positive selection patterns on political interest. Graduates from advantaged backgrounds (A-level holders, early-cohort enrolees, upper PS quintiles) improve their political interest more than the average. Negative selection appears in the partisan support for the Labour party. Among A-level holders the ATT is null, however, the propensity-quintile decomposition picks up a gain among UK graduates from disadvantaged backgrounds, the same population that the Labour vote-proxy ATT captures but the leaning ATT does not. UK graduates from disadvantaged backgrounds carry

[4]East-only sub-panel is unfortunately too small for the bootstrap intervals.

Table 5: Effects of university attainment by propensity to enter university (Zhou–Xie local-IV, Germany)

		Counterfactual averages		Marginal effect at	
Outcome	Sample	On the treated	On the untreated	Low propensity	High propensity
Political interest	All Germany	$+1.07^{***}$	$+1.00^{***}$	$+1.09^{***}$	$+0.99^{***}$
		(0.17)	(0.20)	(0.17)	(0.17)
	West only	$+0.39$	$+0.63^{**}$	$+0.59^{**}$	$+0.43$
		(0.26)	(0.29)	(0.26)	(0.26)
Leans Greens	All Germany	$+0.63^{***}$	$+0.92^{***}$	$+0.71^{***}$	$+0.94^{***}$
		(0.17)	(0.16)	(0.13)	(0.15)
	West only	$+0.48^{**}$	$+0.71^{***}$	$+0.54^{**}$	$+0.74^{***}$
		(0.23)	(0.25)	(0.21)	(0.24)

Notes: Bootstrap SEs in parentheses (500 reps); *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Zhou–Xie local-IV MTE (53); instrument: birth Bundesland; controls: gender, birth year, migrant status, parental education, parental ISEI, HE eligibility. Support trimmed to $[0.02, 0.98]$; estimates on the local-IV latent scale.

Marginal effects (MPRTE) evaluated at the 25th and 75th propensity percentiles. $TT < TUT$ with effects rising in propensity indicates positive selection on gains.

East-only panel and remaining outcomes (party ID, CDU, SPD, Linke) in Appendix Table 19.

the support for the Labour party; UK graduates from advantaged backgrounds do not. This is the inverse of the German MTE pattern on Greens, and reproduces the negative-selection patterns found in the application of the Brand and Xie (8) HTE model for Labour, reported in Appendix Table 14.

A descriptive addition to these propensity estimates is checked against four moderators of the ATT: gender, parental education, migrant or ethnic background, and parental occupational status (AIPW interactions). The Greens shift does not differ across any of the four. Engagement effects are larger among migrants and among graduates from low-parental-ISEI households, consistent with the resource-substitution channel of Lindgren et al. (24), but the direction effect is uniformly distributed across the moderators.

Heterogeneity by field of study (Germany)

The attainment effect varies by the field a person graduates from in terms of engagement and the parties they vote for. In terms of which parties they support the CECT field doesn't moderate the attainment effect significantly (Table 6, Panel A). On the lean outcomes (Table 6, Panel A), high-CECT and low-CECT graduates shift toward Greens by statistically indistinguishable amounts. Comparing the two tracks directly within tertiary (Panel B) yields a significant difference on engagement but a null on partisan direction. However, the composite variable demonstrates that a high-CECT graduate that supports the greens will also vote for the greens (0.96), whereas a low-CECT graduate will only do so about one fifth of the time (0.22). While the political interest does not vary by CECT fields, the decision to vote for the party you support is much more likely for high-CECT graduates. SOEP by CECT field and track demonstrates (Table 6, Panel A)

Table 6: Heterogeneity by field of study (Germany, SOEP)

	Engagement		*Party support*			
Treatment	Political interest	Has party	CDU/CSU	SPD	Greens	Linke
Panel A: Stratified ATT of attainment by field track (vs. non-attainers)						
Tertiary | High-CECT	+0.020	+0.049*	−0.028	+0.060	+0.057*	−0.049*
	(0.036)	(0.026)	(0.041)	(0.041)	(0.033)	(0.028)
Tertiary | Low-CECT	+0.069*	+0.088***	−0.040	+0.051	+0.083***	−0.035
	(0.037)	(0.022)	(0.034)	(0.038)	(0.024)	(0.022)
Panel B: Within-completers contrast (high-CECT vs. low-CECT graduates)						
CECT contrast	+0.069**	+0.067***	−0.034	+0.060	+0.011	−0.011
	(0.035)	(0.023)	(0.039)	(0.038)	(0.032)	(0.028)

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$; bootstrap SEs in parentheses. Estimator: Callaway–Sant'Anna staggered DiD with IPW, replacing the IFE estimator of Hooghe et al. (20) on the same panel to enable the cohort decomposition in Figure 1. Panel A estimates tertiary attainment within each CECT track against track-local never-treated controls (Table 2, Panel B). Panel B reports the high- vs low-CECT contrast within tertiary completers, isolating field content from attainment.

Tables 6 and 7 demonstrate that the ATT is consistent across CECT categories. A benefit of the staggered approach to the the tertiary effect, (Figure 1) demonstrates how this effect is primarily located among the 2005-09 graduates. Both estimator families recover the same all-cohort high-CECT Greens ATT (Tables 6 and 7). The 2005–09 cohort is part of the first cohorts that entered after the Bologna reforms. More importantly, they received their credentials and arrived on the labour market at a crucial time for sustainability issues in the Zeitgeist. The fect estimator used by Hooghe et al. (20) to estimate field effects collapses all the periods and cohorts to in events and the 2005–09 concentrated effect is understated. Field of study *does* predict university's effect on

Table 7: Field of study moderates the vote-anchored partisan effect of tertiary attainment (Hooghe-extension fect, SOEP)

	Greens	SPD	CDU/CSU	Linke
Panel A: vote-anchored composite ATT by field stratum				
High-CECT track	$+0.055^{*}$	-0.024	-0.003	$+0.022$
	(0.030)	(0.021)	(0.025)	(0.014)
Low-CECT track	$+0.018$	-0.008	-0.008	$+0.005$
	(0.020)	(0.021)	(0.020)	(0.018)
High $-$ Low	$+0.037$	-0.016	$+0.005$	$+0.017$
	(0.036)	(0.030)	(0.032)	(0.023)
Panel B: corresponding staggered-DID ATT on the lean outcome (from Table 6, Panel A)				
High-CECT track	$+0.057^{*}$	$+0.060$	-0.028	-0.049^{*}
	(0.033)	(0.041)	(0.041)	(0.028)
Low-CECT track	$+0.083^{***}$	$+0.051$	-0.040	-0.035
	(0.024)	(0.038)	(0.034)	(0.022)
Panel C: lean-to-vote translation ratio (Panel A point estimate $\div$ Panel B point estimate)				
High-CECT track	0.96	—	—	—
Low-CECT track	0.22	—	—	—

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$; bootstrap SEs in parentheses (200 reps). Panel A: Xu et al. (51) IFE on the vote-anchored composite, voter sample, extending Hooghe et al. (20) from lean-only to lean-plus-vote; matrix-completion estimates coincide at every point. Panel B reproduces the lean ATT from Table 6, Panel A. Panel C reports Panel A/Panel B as a descriptive lean-to-vote translation ratio. The all-cohort high-CECT Greens estimates (vote $+5.5$ pp, lean $+5.7$ pp) mask a $+15$ pp effect in the 2005–09 enrolment cohort (Figure 1).

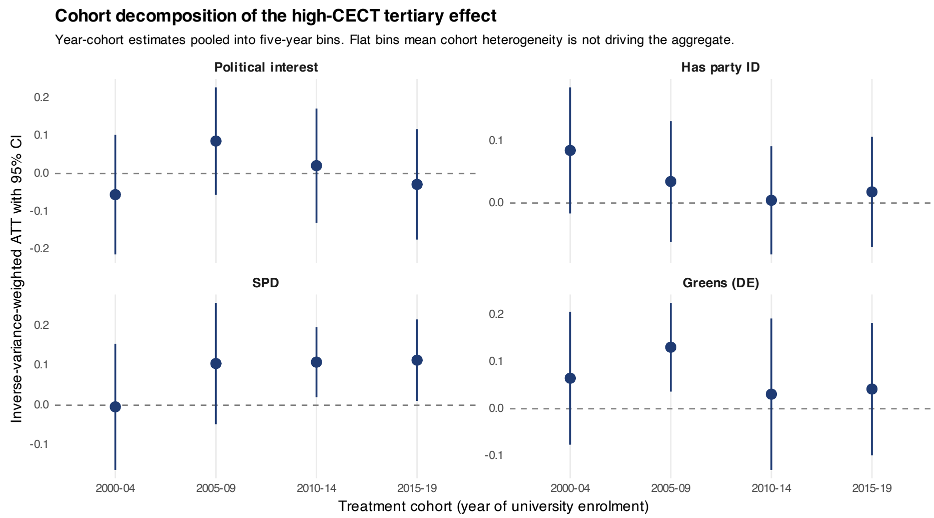

Figure 1: ATTs grouped by years of enrolment

partisanship. Not on the size of the Greens leaning, which is similar across tracks, but on whether leaning towards a party actually manifests in electoral behaviour.

The CECT classification used Hooghe et al. (20) (and in Table 6) assigned science, mathematics, and computing to the high-CECT side, largely due to communicative content. An alternate reading and a more colloquial understanding of a dichotomy between STEM related fields and Liberal arts related fields is estimated in Table 8. Liberal Arts graduates here include the humanities, the arts, the social sciences, teacher training, the caring professions, and law. The liberal arts graduates shift (8%) towards electoral support of the Greens. On the other hand, STEM and Economics graduates do not demonstrate a shift. Liberal Arts graduates also shift toward die Linke in party identification by whopping sixteen percentage points. The Liberal Arts minus STEM difference is statistically significant on die Linke leaning and marginally significant on the Greens vote-proxy. STEM graduates do not carry a detectable partisan effect.

Pre-trends, placebos, and design sensitivity

I run three tests on the staggered difference-in-differences estimates to see if they reflect genuine effects or artifacts of the design. The first is a pre-treatment placebo. I re-estimated the same staggered difference-in-differences specification on political outcomes measured *before* anyone in

Table 8: Liberal Arts vs STEM field-stratified fect estimates (SOEP)

	Greens	SPD	CDU/CSU	Linke
Panel A: Vote-anchored composite (`vote_proxy_X`)				
Liberal Arts	$+0.083^{**}$	-0.027	-0.022	$+0.027^{*}$
	(0.034)	(0.025)	(0.030)	(0.015)
STEM / Economics	$+0.008$	-0.014	$+0.014$	-0.010
	(0.021)	(0.023)	(0.017)	(0.012)
Liberal Arts − STEM	$+0.075^{*}$	-0.013	-0.036	$+0.037^{*}$
	(0.040)	(0.034)	(0.034)	(0.019)
Panel B: Lean (`supports_X`)				
Liberal Arts	$+0.043$	-0.036	-0.129	$+0.161^{**}$
	(0.085)	(0.061)	(0.109)	(0.069)
STEM / Economics	-0.012	$+0.020$	$+0.001$	-0.022
	(0.015)	(0.031)	(0.037)	(0.043)
Liberal Arts − STEM	$+0.055$	-0.056	-0.130	$+0.183^{**}$
	(0.086)	(0.068)	(0.115)	(0.081)

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$ (bootstrap standard errors, 200 replicates). Matrix-completion estimates align with interactive fixed effects (IFE) from Xu et al. (51), focused on voter samples.
Covariates include gender, parental education, migration status, and parental ISEI. Sample sizes: Liberal Arts $n = 1,473$ (603 treated) and STEM/Economics $n = 2,239$ (811 treated). The classification differs from CECT, grouping law with Liberal Arts and science with STEM.

Table 9: Hypotheses and verdicts

	Hypothesis	**Verdict**
H1	Tertiary attainment has a causal effect on political behaviour, independent of pre-adult selection.	Supported. The effect persists under conditioning on pre-treatment outcomes — clearly for partisan direction, modestly for engagement.
H2	University graduation shifts partisan support toward green and left-libertarian parties, not toward the left as a whole.	Supported at the ballot. Graduates' votes move to the Greens (Germany) and Liberal Democrats (UK), not the redistributive centre-left; the SPD gains graduate identification but not votes.
H3	The political return to attainment varies systematically with the likelihood of attaining a degree.	Supported. The return is heterogeneous — positive selection on the Greens in Germany, negative selection on Labour in the UK.
H4	A field's communicative-cultural content shifts partisan behaviour independently of attainment.	Partly supported. Content does not move the size of the partisan leaning, but it governs vote-translation and cohort timing; liberal-arts graduates carry a partisan return STEM graduates do not.

the sample has entered higher education. If the estimator returned an apparent effect in this window — where no treatment has yet occurred — it would mean that future graduates and non-graduates were already on diverging trajectories, and that the post-treatment estimates pick up that prior divergence rather than the effect of the degree itself. None of the placebo estimates are significant at conventional thresholds: there are no detectable pre-trends, and the parallel-trends assumption is not violated.

The second relaxes that assumption rather than testing it, since parallel trends cannot be verified for the post-treatment period itself. I apply the bounded-violation sensitivity analysis of Rambachan and Roth (32), which asks how large a departure from parallel trends the data could tolerate before an estimated effect loses significance. The answer is a breakdown value: the largest post-treatment violation (measured as a multiple of the largest violation actually observed before treatment) under which the effect remains significant. Across all six outcomes each effect survives only so long as any post-treatment departure from parallel trends is no more than half the size of the largest pre-treatment deviation already visible in the data. This is a moderate degree of robustness. A reader who believes post-treatment trends could diverge by more than that should read the staggered estimates as suggestive — and can rely instead on the marginal-treatment-effect results of section , which are identified through the birth-region instrument rather than parallel trends and are therefore untouched by this concern.

The third test involves randomizing the standard errors. I randomly reassign the timing of the educational transition among those eligible for it, producing samples in which any estimated effects are spurious by construction, and re-estimate fifty times. A valid test at the 5% level should reject this fabricated null about 5% of the time. For four of the six outcomes the placebo rejection rate stays near that nominal level (below 8%); for the Greens and the centre-left it runs higher, meaning the conventional standard errors for those two outcomes are somewhat too small and their significance somewhat overstated. Once this is corrected for, the all-Germany Greens effect still clears significance, while the West-Germany treatment-on-the-treated estimate from becomes borderline.

Discussion

Graduating from university does not lead to equal political returns for everyone that attends it. The attainment effects vary by who can graduate or access a degree and what they studied in their degree. When taking into account partisanship that materializes these heterogeneities are also contingent on electoral context of a country.

Interrogating the inequalities of selection itself has resulted in a nuanced picture of tertiary attainment effects. If only the aggregate effects were reported as an event study, treating educational choice "as random" using individual fixed effects to estimate an ATE, I could have reported a single tertiary effect on engagement and a single tertiary effect on partisan direction in each country, and concluded that university produces a modest civic gain and a leftward lean. Treating the propensity score as descriptively informative, rather than a technical tool for robustness, allows for recovering patterns concealed in the aggregate causal estimands.

Whether the individuals who benefit most from a degree are those least likely to have obtained one is theoretically meaningful and highly informative in the selection-versus-socialization debate. The aggregate results in Table 3 show graduates moving toward socially liberal parties of the left in both countries. In Germany, both the SPD and the Greens gain at the level of party leaning (Figure 1), by comparable margins, while support for the CDU/CSU and die Linke remains unaffected. In the UK, the only significant shift at the level of party identification is toward the Liberal Democrats. The Labour effect is statistically indistinguishable from zero, and the Conservative and Green Party effects are too. The two parties that the graduate shift runs through, the Greens in Germany and the Liberal Democrats in the UK, share a profile that distinguishes them from the social democratic mainstream: both sit on the left of center, and both anchor their appeal in social liberalism and cultural openness rather than in economic redistribution. The Greens hold a salient policy anchor in climate and environmental issue ownership that the Liberal Democrats lack, which goes some way to explaining why the German shift is more electorally consequential than the British one, but the underlying graduate preference, for a socially liberal left rather than a redistributive one, is shared across the two settings.

The vote-proxy results in Table 4 reveal a deeper asymmetry between what graduates say their party preference is and what they do at the ballot. In Germany, political support for the Greens translates cleanly into votes, while the SPD support does not: graduates report leaning towards the SPD at the aggregate, but do not turn out for the SPD in elections at a higher rate than non-graduates. In the UK, the asymmetry runs in the opposite direction. Graduates do not report leaning Labour, but they vote Labour at substantially higher rates than non-graduates. The British electoral system, in which the Liberal Democrats and the Greens cannot realistically win most constituencies, forces graduates who lean toward a socially liberal alternative to cast Labour ballots if they want any chance of an outcome they prefer over the Conservatives. The German electoral system, where the Greens can win seats and form governments in their own right, does not impose this kind of strategic cost; the lean and the vote line up.

The likelihood-of-graduating decomposition in § sharpens this cross-country picture rather than parallels it. In Germany, the marginal effect on the Greens rises across the propensity-to-attain distribution, and the counterfactual effect on the untreated exceeds the effect on the treated. Graduates from advantaged backgrounds, those most likely to graduate, drive the Greens shift; less-likely-to-graduate attainers benefit less from the same credential on this outcome. The UK decomposition inverts this on the centre-left. Among A-level holders the Labour effect is null, while the propensity-quintile decomposition recovers a Labour gain concentrated among graduates from disadvantaged backgrounds, the same population the Labour vote-proxy captures but the leaning estimate does not. Where the German result shows advantaged-graduate selection into the socially liberal left, the UK result shows disadvantaged-graduate selection into the traditional left. The Brand-Xie selection-on-observables estimates in Appendix Tables 13 and 14 reproduce these gradients under a stronger identifying assumption; the local-IV identification of § carries them through under a substantially weaker one.

This pattern describes the value-oriented rather than redistributive character of the education cleavage. Graduates concentrate in socially liberal parties of the left, where the electoral system makes them viable alternatives, the Greens in Germany and the Liberal Democrats in the

UK, at the level of party identification. The economically redistributive traditional center-left does not attract graduate identification in the UK, and does not attract graduate votes in Germany, even where it does attract graduate identification there. The finding speaks to Piketty's "Brahmin left" thesis that the educational divide in voting is driven primarily by green and left-libertarian parties rather than by the social democratic family (1). Bourdieu's account of the "new petite bourgeoisie" (4) fits with this pattern; the professional class reproduces its own culturally tolerant and economically libertarian attitudes through higher education. The German positive selection on the Greens and the UK negative selection on Labour are two faces of this same dynamic. Advantaged graduates flow into socially liberal alternatives where the electoral system makes them viable, and out of the dominant center-left where it does not.

Within the university gate, the field-of-study story has changed shape. The within-completers contrast between high-CECT and low-CECT graduates on the size of the Greens leaning is null (Table 6, Panel B): the partisan-direction effect of attending university is not significantly larger for graduates of communicative-cultural fields than for graduates of technical and economic ones. Where the field of study does matter for partisan direction is at the ballot. The vote-anchored composite in Table 7 shows that a high-CECT graduate who leans Greens turns out to vote Greens roughly nineteen times in twenty; a low-CECT graduate who reports the same lean does so about one time in five. The communicative-cultural field channel is not about generating a Greens preference, since low-CECT graduates report Greens lean at comparable rates, but about anchoring that preference in ballot behavior. The cohort decomposition in Figure 1 adds the second piece. The high-CECT Greens effect is concentrated in the 2005-09 enrolment cohort, the first to complete under the post-Bologna structure, and the cohort that arrived on the German labor market as climate and sustainability issues took up a central place in the public political conversation. Aggregate estimates of tertiary effects on political behavior that are not sensitive to period risk, extrapolating a time-bounded causal effect, e.g. Hooghe et al. (20). In short, the field of study does not predict the size of support for the Greens leaning, but it does predict whether that support turns into a vote.

The contrast becomes sharper under the Simon-Scott Liberal Arts versus STEM classification (Table 8). The Hooghe-Werfhorst classification assigns the natural sciences, mathematics, and computing to the high-CECT side on the grounds that these subjects develop communicative skills as part of their training. The Liberal Arts versus STEM contrast assigns them instead to the STEM side, reflecting the substantive content of the disciplines rather than the competencies they happen to develop. Under this alternative grouping, Liberal Arts graduates (the humanities, the arts, the social sciences, teacher training, the caring professions, law) shift toward the Greens at the ballot by roughly eight percentage points, and toward the Linke in party identification by sixteen. STEM and Economics graduates show no detectable shift on any of the four parties at either the leaning or the vote-anchored level. The Liberal Arts minus STEM contrast is significant on Linke leaning and marginally significant on the Greens vote-proxy. The Hooghe-Werfhorst split picks up part of the field contrast through the communicative-skill content of the natural-science curriculum, but obscures the disciplinary boundary that the substantive-content grouping puts where it belongs.

Several mechanisms account for why the hard sciences and mathematics carry no detectable partisan return where the humanities, social sciences, and caring professions do. The first is curricular content. The Liberal Arts disciplines reflexively examine social structures, contested foundations of political arrangements, and the cultural and distributional stakes of public life as part of their object of study. STEM disciplines treat their object of study (nature, computation, formal structures) as analytically prior to those arrangements; whatever communicative skill a physics or mathematics degree develops is developed in the service of a non-political epistemology that does not, in itself, position the graduate within the political space. The second is occupational destination. STEM graduates sort disproportionately into private-sector technical roles, engineering, finance, and IT. Liberal Arts graduates sort disproportionately into education, public-sector administration, the cultural professions, and the caring sector, occupational locations that Oesch (27) identifies as the sociocultural professional fraction whose work logic generates the value-liberal political orientations the Brahmin-left literature documents. The third is possibly prior orientation. People who choose mathematics or physics may already be more politically agnostic at enrolment than those who choose history or sociology. The null partisan effect among completers is, however, hard to reduce to prior disposition alone, since the Liberal Arts effect emerges among completers from broadly similar pre-treatment trajectories. The conjunction of curricular content and occupational sorting makes the STEM null finding a substantive contrast rather than a measurement artefact. It is not that university education produces partisan effects and STEM is just lower in dosage; it is that the partisan return to higher education runs specifically through the Liberal Arts and the occupational positions they lead to.

The structural-position moderation in § produces a clean asymmetry between the engagement and the direction effects. The treatment effects on the Greens do not vary by any of the four moderators tested (gender, parental education, migrant or ethnic background, parental occupational status). The effect is uniformly distributed across the structural variables that typically condition political socialization. In participation, the effect of political interest is greater among graduates of migrant backgrounds and among graduates of low-SES households, consistent with the channel of resource-substitution Lindgren et al. (24). When there is less room for socializing, civic participation, and political articulation during adolescence, the university supplies this gap, and the marginal engagement gains are therefore larger among these people. The credential channel on engagement is, therefore, compensatory for structural disadvantage, where the political directional returns are not.

Across these layers a single reorientation suggests itself. The credential on its own produces a modest engagement gain in both countries and a partisan lean toward the socially liberal wing of the party system. But the credential by itself does not discriminate well among graduates. The direction effect is uniformly distributed across the structural moderators, the propensity-varying gradient runs in opposite directions on the two relevant parties across the two countries, and the high-CECT versus low-CECT contrast within the German tertiary sample is null on the size of the Greens leaning. What the credential does not discriminate, the curriculum does. The Liberal Arts versus STEM contrast on the same panel produces a partisan return on the Greens at the ballot and on the Linke at the leaning level that the credential contrast cannot, and the lean-to-vote anchoring asymmetry between high-CECT and low-CECT graduates shows that the field channel operates on what graduates do with a political preference rather than on whether they form one. Brady et al. (6) decomposed the social-status-participation link through resources, demonstrating that civic skills and recruitment networks drive participation independently of political interest. The present paper arrives at the same conclusion from the opposite direction. The political return to higher education is governed less by the credential and more by the specific communicative, cultural, and discursive disciplines that Verba et al. (47) identifies as the curricular site where participatory skills are cultivated. Future work on education and democracy should disaggregate the credential from the curriculum, attend carefully to pre-treatment heterogeneity, and resist treating university education as a uniform civic treatment when its political consequences are sharply conditional on what is taught and to whom.

Three limits define the present design. The first concerns parallel trends. The bounded-violation sensitivity test holds at the five-year post horizon under a breakdown value of $\bar{M} = 0.5$. A reader unwilling to assume parallel trends to that tolerance should treat the staggered DiD estimates as suggestive and rely on the instrumental-variable evidence for the German heterogeneity results. The second concerns the random-reassignment placebo, which shows mild over-rejection on the Greens and centre-left outcomes; the all-Germany Greens effect survives size correction, but the within-West local-IV estimate becomes borderline, and I treat it as such. The region at birth (Bundesland) instrument works, unfortunately the same cannot be done for the UK given data constraints. the UK heterogeneity layer rests instead on the non-parametric three-way decomposition. A compulsory-schooling instrument in a comparable national context within the same period is the natural extension of this paper. In a forthcoming paper, I move from field classification to course-credit exposure, using credit-by-field as a continuous treatment that varies within degree programs and permits a more precise response-dosage estimation of curricular exposure effects on political outcomes.

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Appendix A. Estimators and identifying assumptions

This section provides the formal specifications of the estimators used in the article; see section .

A.1 Staggered difference-in-differences

The approach follows Callaway et al. (9) and is used for the SOEP aggregate effect on engagement and party-support outcomes (section (a)) and for the field-of-study heterogeneity (section (d)).

Let $G_i \in \{2000, \ldots, 2020\} \cup \{0\}$ denote the calendar year of respondent i's first reported tertiary enrolment, with $G_i = 0$ for never-treated respondents. The group-time average treatment on the treated is

$$\mathrm{ATT}(g,t) = E\big[Y_{it}(1) - Y_{it}(0) \,\big|\, G_i = g\big], \quad t \geq g. \tag{1}$$

Under conditional parallel trends and no-anticipation (9, Assumptions 3–4), $\mathrm{ATT}(g,t)$ is identified by the inverse-probability-weighted estimator

$$\widehat{\mathrm{ATT}}(g,t) = E\left[\left(\frac{1\{G_i = g\}}{P(G_i = g)} - \frac{p_g(X_i)\,(1 - D_i^g)}{\big(1 - p_g(X_i)\big)\,P(G_i = g)}\right)\big(Y_{it} - Y_{i,g-1}\big)\right], \tag{2}$$

where $p_g(X) = P(G_i = g \mid X_i, G_i \in \{g, 0\})$ is the generalised propensity score, X_i contains gender, parental education, migrant background, and year of birth, and $D_i^g = 1\{G_i = g\}$. The overall ATT is the simple aggregation across groups and post-treatment periods,

$$\mathrm{ATT}_{\text{overall}} = \frac{1}{|\mathcal{T}|} \sum_{(g,t) \in \mathcal{T}} \mathrm{ATT}(g,t), \tag{3}$$

with $\mathcal{T} = \{(g,t) : t \geq g\}$. Inference uses a 999-replicate multiplier bootstrap. Identification rests on conditional parallel trends between treated cohort g and the never-treated comparison group at every post-treatment t, no anticipation in pre-treatment periods, and the standard overlap condition $0 < p_g(X) < 1$.

The no-anticipation assumption is testable through the pre-treatment event-study coefficients $\widehat{\mathrm{ATT}}(\tau)$ at $\tau < 0$, which should be jointly indistinguishable from zero. In the present application the pre-trends are flat for partisan-direction outcomes and the vote-or-lean composite, but show significant pre-treatment ATTs for engagement outcomes. HonestDiD sensitivity (32) is reported on every outcome to quantify the violation of parallel trends under which the post-treatment ATT remains distinguishable from zero.

A.2 Interactive fixed effects

The approach follows Xu et al. (51) and is used for the UK aggregate effect on all outcomes (section (a)) and the vote-or-support behavioural extension in both panels (section (e)).

The untreated outcome is modelled as

$$Y_{it}(0) = \alpha_i + \xi_t + \lambda_i^\top f_t + X_{it}^\top \beta + \varepsilon_{it}, \tag{4}$$

where α_i is a person fixed effect, ξ_t a wave fixed effect, $f_t \in R^r$ a vector of latent time-varying factors, $\lambda_i \in R^r$ the person-specific factor loadings, X_{it} contains gender, ethnicity (UK) or migrant background (Germany), parental education, and year of birth, and ε_{it} is the error term. Factor rank r is selected by cross-validation; $r = 0$ recovers two-way fixed effects without latent-factor structure. The counterfactual $\widehat{Y}_{it}(0)$ is imputed under nuclear-norm regularisation. Treatment D_{it} is absorbing: $D_{it} = 1$ from the first wave a respondent is observed in the treated state. The counterfactual for treated observations is imputed by fitting (4) on the never-treated subsample and predicting the treated sample under the same parameters. The ATT, with $\mathcal{T}_1$ the set of treated observations with at least one pre-treatment observation in the UK and at least five in Germany, is

$$\widehat{\mathrm{ATT}} = \frac{1}{|\mathcal{T}_1|} \sum_{(i,t)\in\mathcal{T}_1} \big(Y_{it} - \widehat{Y}_{it}(0)\big). \tag{5}$$

Standard errors are obtained from a 200-replicate bootstrap. The covariate vector X_{it} enters the outcome model: covariates adjust for observed confounders in the counterfactual prediction, while the latent factor term $\lambda_i^\top f_t$ absorbs unobserved time-varying confounders. This is consequential because earlier work has found that unobserved time-varying confounders can account for the bulk of the estimated effect (e.g. 53).

Identification requires: (i) individual and time fixed effects absorb all time-invariant variation and common shocks; (ii) the latent factor absorbs unobserved time-varying confounders correlated with treatment; and (iii) the counterfactual fitted on never-treated respondents extrapolates correctly to treated respondents absent treatment. The third assumption is tested in a placebo equivalence test that holds the cross-validated factor rank fixed and reserves event-times $\tau \in \{-2, -1\}$ out of the ATT computation; the parallel-trends assumption is supported if the placebo ATT on the reserved periods is statistically indistinguishable from zero ($p > 0.10$).

A.3 Marginal treatment effect

The approach follows Zhou and Xie (54) and is used for the Germany heterogeneity-by-likelihood layer (section (b)).

Treatment selection is modelled through a latent-index threshold-crossing rule (16). Each respondent is characterised by two quantities. The propensity score,

$$P(Z_i, X_i) = P(D_i = 1 \mid X_i, Z_i)\,, \tag{6}$$

is a scalar summary of observed characteristics X_i (gender, year of birth, migrant background, parental education, parental occupational status, Abitur eligibility) and the local instrument Z_i (Bundesland of birth, categorical). The latent resistance quantile is

$$U_i = F_{V|X}(V_i) \sim \mathrm{Uniform}(0, 1), \tag{7}$$

where V_i is unobserved individual resistance to treatment. The selection rule is then

$$D_i = 1\{P(Z_i, X_i) > U_i\}\,: \tag{8}$$

a respondent is treated if and only if observed propensity exceeds unobserved resistance.

The original MTE of Heckman and Vytlacil (16), $\mathrm{MTE}(x,u) = E[Y_i(1) - Y_i(0) \mid X_i = x, U_i = u]$, conditions on the full covariate vector and is high-dimensional. Zhou and Xie (54) reparameterise by replacing X_i with its scalar propensity-score summary,

$$\widetilde{\mathrm{MTE}}(p,u) = E\big[Y_i(1) - Y_i(0) \,\big|\, P(Z_i, X_i) = p,\ U_i = u\big]\,. \tag{9}$$

Because (P, U) jointly determines treatment status, $\widetilde{\mathrm{MTE}}$ captures all of the heterogeneity that contributes to selection bias.

Three estimands emerge from weighted integrals of $\widetilde{\mathrm{MTE}}$ over the joint distribution of (P, U). The ATE under random assignment is

$$\mathrm{ATE} = \int_0^1 \int_0^1 \widetilde{\mathrm{MTE}}(p,u)\, dF_{P,U}. \tag{10}$$

The ATT, the expected effect among respondents who actually received treatment ($u < p$), is

$$\mathrm{ATT} = \int_0^1 \int_0^p \widetilde{\mathrm{MTE}}(p,u)\, dF_{P,U|D=1}. \tag{11}$$

The treatment effect on the untreated (TUT), the counterfactual effect among those with $u \geq p$, is

$$\mathrm{TUT} = \int_0^1 \int_p^1 \widetilde{\mathrm{MTE}}(p,u)\, dF_{P,U|D=0}. \tag{12}$$

The contrast $\mathrm{ATT} - \mathrm{TUT}$ has a direct interpretation: $\mathrm{ATT} > \mathrm{TUT}$ indicates positive selection on gains, while $\mathrm{ATT} < \mathrm{TUT}$ indicates negative selection on gains (8).

The marginal policy-relevant treatment effect at propensity level p is

$$\mathrm{MPRTE}(p) = \widetilde{\mathrm{MTE}}(p,p), \tag{13}$$

the expected effect among respondents at the margin of indifference, those an incremental policy change at level p would move in or out of treatment. $\mathrm{MPRTE}(p)$ decomposes into an observed-heterogeneity component (variation in p at fixed u) and an unobserved-gain component (variation in u at fixed p). Negative selection on gains is operationalised as $\mathrm{ATT} > \mathrm{TUT}$ with $\mathrm{MPRTE}(p)$ declining in p; the reverse pattern indicates positive selection at the margin.

Identification rests on: (i) the exclusion restriction, Z_i affects political outcomes only through D_i conditional on X_i; (ii) instrument relevance, Z_i shifts the propensity score conditional on X_i; and (iii) additive separability of $Y_i(d)$ in X_i and U_i, plus standard local-IV support conditions on the joint density of (P, U). The exclusion restriction is plausible in the German sample because Bundesland of birth is time-invariant and affects regional university access and exposure to academic trajectories; the relevance condition is verified by a likelihood-ratio test on the sixteen Bundesland indicators ($\chi^2_{15} = 116.22$, $p < 10^{-6}$). The parallel UK pipeline using a tuition-fee-shock instrument fails the relevance test ($\chi^2_3 = 2.62$, $p = 0.45$), consistent with the price-inelasticity of UK enrolment documented in Murphy et al. (26); the stratification-based heterogeneity estimation (section (b)) is used for the UK instead.

A.4 Augmented inverse-probability-weighted interaction

The augmented inverse-probability-weighted (AIPW) interaction estimator is used for the heterogeneity-by-structural-predictors layer in both panels (section (c)). For each moderator $M \in \mathcal{M}$ and outcome Y, the within-stratum ATT is estimated separately on the strata $M = 0$ and $M = 1$:

$$\widehat{\mathrm{ATT}}_{M=m} = E\left[\mu_1(X) - \mu_0(X) + \frac{D - e(X)}{e(X)\big(1 - e(X)\big)}\big(\Delta Y - \mu_D(X)\big) \,\Big|\, M = m\right], \tag{14}$$

where $\mu_d(X)$ is the outcome regression of within-person change $\Delta Y = Y_{\text{post}} - Y_{\text{pre}}$ on X in stratum $\{D = d\}$, $\mu_D(X) = D \cdot \mu_1(X) + (1 - D) \cdot \mu_0(X)$, and $e(X) = P(D = 1 \mid X)$ is the propensity score. The moderation contrast is

$$\Delta_M = \widehat{\mathrm{ATT}}_{M=1} - \widehat{\mathrm{ATT}}_{M=0}, \tag{15}$$

with standard errors from a 200-replicate non-parametric bootstrap. The estimator is doubly robust (34): consistent for $\mathrm{ATT}_{M=m}$ under correct specification of either μ_d or $e(X)$. The moderator set is gender, high parental education (mother and father separately), high parental occupational status, and ethnicity/migrant background.

Appendix B. Sample, propensity scores, and sensitivity

Table 10: Effective sample size by estimation design and treatment status

	Treated (Tertiary)			Control (No Tertiary)		
Outcome	N_{HTE}	N_{DRDID}	Total	N_{HTE}	N_{DRDID}	Total
Panel A: SOEP (Germany) — Tertiary treatment						
Pol. Interest	9,361	1,593	10,794	5,864	1,237	16,572
Left Party	3,883	823	10,794	560	154	16,572
New-Left/Greens	3,883	823	10,794	560	154	16,572
Centre-Left (SPD)	3,883	823	10,794	560	154	16,572
Panel B: UKHLS (UK) — Tertiary treatment						
Pol. Interest	5,956	2,066	13,566	1,497	552	23,136
Left Party	2,436	1,065	13,566	206	104	23,136
New-Left/Greens	2,436	1,065	13,566	206	104	23,136
Centre-Left (Lab)	2,436	1,065	13,566	206	104	23,136

	High CECT			Low CECT		
Outcome	N_{HTE}	N_{DRDID}	Total	N_{HTE}	N_{DRDID}	Total
Panel C: SOEP — CECT within tertiary completers						
Pol. Interest	999	168	534	3,382	528	1,819
Left Party	419	45	534	1,092	158	1,819
New-Left/Greens	419	45	534	1,092	158	1,819
Centre-Left (SPD)	419	45	534	1,092	158	1,819
Has Party	1,102	262	534	3,486	840	1,819
Panel D: UKHLS — CECT field preference ITT (not restricted to completers)						
Pol. Interest	979	290	541	1,462	432	866
Left Party	542	99	541	703	135	866
New-Left/Greens	542	99	541	703	135	866
Centre-Left (Lab)	542	99	541	703	135	866
Has Party	979	290	541	1,462	426	866

Notes: N_{HTE} counts person-waves with valid outcome and matched pre-treatment outcome (pre-window, ages 15–20). N_{DRDID} counts unique persons observed in both pre- and post-periods. Total is all persons in the relevant comparison group within the main sample (born $\geq$ 1980). SOEP CECT uses completion-based field of degree among tertiary graduates; UKHLS CECT uses field preference from the youth questionnaire at age 16–17, intention-to-treat (not restricted to completers).

Table 11: Tertiary propensity-score model progression

Model	N	**AUC**	R^2	**ΔAUC**
SOEP (model-selection diagnostic, polint_pre as candidate):				
M1: Demographics	7,388	0.850	0.306	—
M2: + polint_pre	7,388	0.856	0.316	$+0.007$
M3: + polint_pre + ISEI	7,388	0.863	0.331	$+0.007$
UKHLS (model-selection diagnostic):				
Parental education	5,846	0.573	0.020	—
+ Gender	5,846	0.593	0.026	$+0.019$
+ Migration	5,846	0.603	0.027	$+0.011$
+ Birth year	5,846	0.603	0.029	$+0.000$
+ A-Level	3,738	0.853	0.236	$+0.250$
+ polint_pre	3,738	0.856	0.248	$+0.002$
Headline propensity score (Brand–Xie pure form, polint_pre in outcome equation):				
SOEP (14 cov.)	4,199	**0.777**	—	—
UKHLS (12 cov.)	3,227	**0.743**	—	—

Notes: The top two blocks report the model-selection diagnostics that motivate the Brand–Xie pure-form choice: pre-treatment political interest adds ΔAUC < 0.01 to the demographic PS in SOEP and $+0.002$ in UKHLS once the academic-track variable (Gymnasium / A-level) is included, while adding A-Level alone lifts the UK AUC from 0.60 to 0.85. Pre-treatment political interest is therefore placed in the outcome equation rather than the propensity score (Brand–Xie pure form, 8). The bottom block reports the headline PS AUC. SOEP (14 cov.): universal core + Gymnasium, ISEI + missing flag, cob_4, cob_8. UKHLS (12 cov.): universal core + ethn_3, migr_gen_1, migr_gen_4.

Table 12: CECT propensity-score model progression (field-choice, within tertiary)

Model	N	AUC	R^2	ΔAUC
SOEP (forward progression, pre-treatment panel ages 10–24):				
M1: Female only	1,238	0.680	0.092	—
M2: + yborn + migrant	1,238	0.694	0.094	$+0.014$
M3: + gymnasium	1,238	0.693	0.094	-0.001
M4: + par_edu_std	1,177	0.697	0.095	$+0.003$
M5: + relig_att (selected PS)	**924**	**0.719**	**0.114**	$+0.022$
Sensitivity: political variables added to the PS				
S1: + polint_pre	924	0.718	0.115	-0.001
S2: + left_pre	708	0.759	0.151	$+0.041$
Headline CECT propensity score:				
SOEP (7 cov.)	276	0.710	0.065	—
Lasso (1 covariate selected)	1,238	0.651	—	—
Unpenalised (12 covariates)	1,238	0.708	—	—
UKHLS (lasso-vs-theory diagnostic):				
Theory-driven (6 cov.)	1,409	0.725	0.077	—
Lasso (λ_{1se}, 2 cov.)	1,409	0.702	—	-0.023

Notes: Logit predicting Pr(high-CECT) among tertiary graduates (SOEP) or among all youth with a stated field preference (UKHLS, ITT design). The top block reports the model-selection progression on the SOEP retrospective field-of-completion design. The sensitivity block confirms that polint_pre adds nothing to the PS while left_pre adds ΔAUC $+0.041$; both nonetheless enter the outcome equation rather than the PS to avoid conditioning on a lagged version of the outcome. The headline SOEP CECT logit uses female, yborn, migrant, gymnasium, fedu3, medu3, relig_att ($N = 276$, AUC 0.710); the UKHLS headline logit uses female, yborn, migrant, fedu3, medu3, has_alevel ($N = 1{,}409$, AUC 0.725, falling to 0.702 under lasso shrinkage to female + parental NS-SEC). The substantive direction is robust to the lasso-vs-theory choice.

Table 13: Full HTE results: SOEP (born $\geq$ 1980)

Design	Outcome	N	ME	SE	β_3	β_3 SE	Pattern
Design A: Tertiary vs Non-Tertiary (with Y_{pre})							
A	Pol. Interest (SD)	27,478	+0.079	0.040	−0.056	0.090	No pattern
A	Left Party	5,939	−0.111	0.022	+0.214	0.053	**Pos. sel.*****
A	Has Party Pref	27,131	−0.054	0.020	−0.092	0.052	Neg. sel.*
A	Centre-Left (SPD)	5,976	−0.093	—	+0.051	—	Pos. sel.
A	New-Left/Greens	5,976	−0.112	—	+0.215	—	Pos. sel.**
Design B: CECT within Tertiary (field-choice PS, raw scale)							
B	Pol. Interest	839	−0.083	—	+0.172	—	Pos. sel.
B	Left Party	1,460	+0.201	0.067	−0.041	0.200	No pattern
B	Has Party Pref	838	−0.266	—	+0.798	—	Pos. sel.
B	Centre-Left (SPD)	353	−0.071	—	−0.017	—	No pattern
B	New-Left/Greens	353	+0.036	—	+0.291	—	Pos. sel.
Gender (left–right position)							
A	Men	—	—	—	−0.643	—	Neg. sel.
A	Women	—	—	—	−0.212	—	Neg. sel.
Voting (Bundestag elections, with Y_{pre})							
A	Voted 2009	1,046	+0.160	—	−0.210	—	Neg. sel.
A	Voted 2013	3,456	+0.106	—	−0.158	—	Neg. sel.

Notes: Cross-sectional IPTW interaction estimates (not the within-person DRDID of the main text). ME = main effect at PS $= 0$; β_3 = treatment $\times$ PS interaction. Design A uses the headline 14-covariate tertiary PS (Brand–Xie pure form: female, fedu3, medu3, migrant, yborn, relig_att, nphh_pre, hhinc_pre, kidsn_pre, par_isei, par_isei_missing, gymnasium, cob_4, cob_8) plus a matched Y_{pre} in the outcome equation (polint_pre for engagement; party_pre for Has Party; centreleft_pre / newleft_pre for SPD and Greens; left_pre for Left Party). Design B uses the 7-covariate CECT field-choice PS (female, yborn, migrant, gymnasium, fedu3, medu3, relig_att). Pol. Interest is in SD units in Design A and in raw 1–4 Likert units in Design B. The Left-Party β_3 is positive: graduates from advantaged backgrounds show the largest within-bloc shift; the aggregate Left-Party DRDID is a precisely-estimated null (**??**).
$^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.

Table 14: Full HTE results: UKHLS (born $\geq$ 1980)

	Analysis	N	ME (SE)	INT (SE)	p	Sel
A1: Tertiary vs Non-Tertiary (with Y_{pre})						
1	Pol. Interest (SD)	18,741	$+0.123$ (0.049)	$+0.214$ (0.076)	0.005	**(+)****
2	Has Party (SD)	19,641	$+0.226$ (0.057)	$+0.091$ (0.087)	0.298	(+)
3	Voted (SD)	7,406	$+0.395$ (0.110)	$+0.200$ (0.168)	0.233	(+)
4	Centre-Left (Labour)	6,663	$+0.023$ (0.129)	-0.114 (0.181)	0.530	($-$)
5	New-Left (composite)	6,663	-0.162 (0.107)	$+0.139$ (0.143)	0.330	(+)
6	Greens (alone)	6,663	-0.043 (0.144)	-0.002 (0.203)	0.991	0
F1: Party-specific HTE (Tertiary, with polint_pre)						
7	Labour	7,920	$+0.175$ (0.070)	-0.302 (0.094)	0.001	**($-$)*****
8	LibDem	7,920	-0.132 (0.037)	$+0.202$ (0.053)	<0.001	**(+)*****
9	SNP/PC	7,920	-0.021 (0.031)	$+0.051$ (0.038)	0.178	(+)
10	Greens (alone)	7,920	$+0.010$ (0.026)	$+0.005$ (0.036)	0.885	0
11	Conservative	7,920	$+0.018$ (0.057)	-0.055 (0.078)	0.481	($-$)
A6: CECT within Tertiary (field-choice PS, ITT)						
12	Pol. Interest	2,430	$+0.581$ (0.137)	-0.583 (0.291)	0.045	**($-$)***
13	Has Party	2,344	$+0.033$ (0.120)	-0.129 (0.261)	0.621	($-$)
14	Voted	1,549	$+0.029$ (0.157)	$+0.111$ (0.340)	0.743	(+)
15	Centre-Left (Labour)	815	$+0.056$ (0.136)	-0.002 (0.280)	0.996	0
16	New-Left (composite)	815	-0.305 (0.161)	$+0.732$ (0.343)	0.033	**(+)***
17	Greens (alone)	815	-0.219 (0.224)	$+0.667$ (0.493)	0.176	(+)

Notes: Cross-sectional IPTW interaction estimates. All samples born ≥ 1980; outcomes z-scored (SD units) unless noted. Sel: (+) positive, ($-$) negative, $0 = |\text{INT}| < 0.05$. Tertiary PS (12 cov.): female, fedu3, medu3, migrant, yborn, relig_att, nphh_pre, hhinc_pre, kidsn_pre, ethn_3, migr_gen_1, migr_gen_4. CECT field-choice PS (4 cov.): female, yborn, migrant, has_alevel. The party-specific block (rows 7–11) uses an extended PS with polint_pre, for parity with the SOEP party decomposition. Labour shows the canonical negative-selection pattern; LibDem moves in the opposite direction; Conservative is a flat null. $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.

Table 15: Parallel-trends pre-tests (pre-window ages 15–20, born $\geq$ 1980)

Test	Obs	Persons	p	Result
SOEP — political interest:				
(A) Unconditional Wald	16,592	5,494	0.294	Fails to reject
(B) Conditional (+ demog.)	12,100	—	0.116	Fails to reject
(C) Person FE + year FE	—	—	0.580	Fails to reject
SOEP — left-party support:				
(A) Unconditional Wald	3,196	1,273	<0.001	Rejects
(B) Conditional (+ demog.)	2,405	—	<0.001	Rejects
(C) Person FE + year FE	—	—	0.218	Fails to reject
UKHLS — political interest:				
(A) Unconditional Wald	33,165	10,575	0.002	Rejects
UKHLS — left-party support (strict):				
(A) Unconditional Wald	8,791	3,265	0.419	Fails to reject

Notes: Pre-period defined as ages 15–20, applied uniformly to treated and control respondents. Joint Wald tests of year-by-year treatment–time interactions, clustered by person. (A) unconditional; (B) conditional on female, fedu3, medu3, migrant, gymnasium; (C) person + year FE. "Fails to reject" at $p > 0.05$; "Rejects" at $p < 0.05$. SOEP Political Interest passes all three specifications. SOEP Left-Party rejects the unconditional and conditional Wald tests but passes the within-person FE specification ($p = 0.218$): the rejection on the marginal tests reflects baseline differences in cohort means between future-tertiary and future-non-tertiary respondents (61% vs 65% support left at age 17–20) rather than differential pre-trends. UKHLS Political Interest rejects the within-period Wald test, consistent with the BHPS-to-UKHLS transition in political-interest question wording. UKHLS Left Party passes cleanly. The within-person test is the relevant identification check for the within-person DRDID of the main text and is satisfied for every headline outcome where it is computable.

Table 16: Sample attrition: from raw panel to DiD estimation sample

	SOEP		UKHLS	
Filter step (cumulative)	N	%	N	%
Person-level cohort				
Total persons in panel	156,646	100.0	84,986	100.0
Tertiary or upper-secondary completers	102,987	65.7	79,493	93.5
Born $\geq$ 1980 (main sample, persons)	27,366	17.5	38,038	44.8
Pre-treatment baseline (age 15–20)				
With non-missing `polint_pre`	18,285	11.7	15,842	18.6
With non-missing `left_pre`	3,608	2.3	8,382	9.9
DiD samples (pre age 15–20 + post age 25+, persons)				
Political interest	2,718	9.9	3,440	9.0
of whom tertiary	1,061	3.9	1,714	4.5
of whom upper secondary	1,657	6.1	1,726	4.5
Left-party support	805	2.9	1,369	3.6
Has-party preference	3,360	12.3	3,537	9.3
Three-period balanced panel (political interest)				
Persons at all three age bands	1,022	3.7	733	1.9

Notes: Cumulative person-level filters. Rows below the main-sample line express N as % of the main sample (27,366 in SOEP, 38,038 in UKHLS). The pre-treatment baseline rows show how many main-sample persons have the corresponding pre-treatment value at age 15–20; the DiD-sample rows show how many additionally have a post-treatment outcome observation at age 25+. The Has-Party DiD sample is larger than the Political-Interest sample because `party_pre` is asked of all adult respondents, whereas the Likert `polint_pre` is asked less frequently in the youth questionnaire.

Table 17: DiD-sample selectivity vs main sample (demographic comparison)

	N	%Trt	%F	BYr	Fed	%Mig	Gym/AL
Panel A: SOEP							
sample_main (universe)	27,366	39.4	51.8	1989.4	2.09	45.0	44.4
DiD: political interest	2,718	39.0	51.4	1988.3	2.13	6.7	43.8
DiD: left party	805	58.9	49.8	1987.8	2.25	4.8	44.0
DiD: has party	3,360	33.8	50.5	1988.4	2.12	8.2	43.5
DiD: New-Left/Greens	805	58.9	49.8	1987.8	2.25	4.8	44.0
DiD: Centre-Left (SPD)	805	58.9	49.8	1987.8	2.25	4.8	44.0
Panel B: UKHLS							
sample_main (universe)	38,038	35.7	52.3	1990.1	1.77	17.3	33.9
DiD: political interest	3,440	49.8	56.9	1989.3	1.86	5.7	38.6
DiD: left party	1,369	64.1	53.8	1989.0	1.63	4.3	44.6
DiD: cultural left	1,369	64.1	53.8	1989.0	1.63	4.3	44.6
DiD: has party	3,537	49.4	56.9	1989.1	1.85	5.6	39.9
DiD: New-Left/Greens	1,369	64.1	53.8	1989.0	1.63	4.3	44.6
DiD: Centre-Left (Labour)	1,369	64.1	53.8	1989.0	1.63	4.3	44.6

Notes: Person-level demographic profile of each DiD sample relative to the main analytical sample (`sample_main`). %Trt = % tertiary-treated. %F = % female. BYr = mean birth year. Fed = mean father's education (raw 1–3 scale); for UKHLS, mean parental NS-SEC class instead. %Mig = % migration background. Gym/AL = % Gymnasium track (SOEP) or A-level (UKHLS). Migrant-background respondents are under-represented in the DiD samples relative to the universe (45.0% vs 5–8% in SOEP; 17.3% vs 4–6% in UKHLS). Tertiary completers are over-represented in the party-preference DiD samples (+19.5 pp in SOEP, +28.4 pp in UKHLS) because the within-person pre-to-post observation requirement is satisfied more often for graduates with denser adult panel coverage. The DRDID estimator addresses selection on the same observables; selectivity on unobservables remains a threat addressed by the parallel-trends pre-test (Table 15).

Table 18: Pre/post window sensitivity for headline ATTs

Spec	N	ATT	SE	95% CI	Sig
SOEP — political interest (z-scored, SD units)					
Headline (pre 15–20, post 25+)	2,718	+0.132	0.048	[+0.038, +0.226]	**
Tight pre (16–19, post 25+)	2,474	+0.136	0.049	[+0.040, +0.233]	**
Wide pre (14–22, post 25+)	3,323	+0.086	0.046	[−0.005, +0.177]	—
Late post (15–20, post 28+)	1,541	+0.225	0.065	[+0.098, +0.352]	**
Much later (15–20, post 30+)	1,059	+0.248	0.073	[+0.104, +0.391]	**
SOEP — has-party preference (0/1)					
Headline (pre 15–20, post 25+)	3,360	+0.150	0.023	[+0.104, +0.195]	**
Tight pre (16–19, post 25+)	3,027	+0.155	0.023	[+0.109, +0.201]	**
Wide pre (14–22, post 25+)	4,212	+0.130	0.022	[+0.086, +0.174]	**
Late post (15–20, post 28+)	1,870	+0.123	0.032	[+0.061, +0.185]	**
Much later (15–20, post 30+)	1,246	+0.154	0.036	[+0.083, +0.225]	**
UKHLS — political interest (z-scored, SD units)					
Headline (pre 15–20, post 25+)	3,440	+0.051	0.041	[−0.030, +0.131]	—
Tight pre (16–19, post 25+)	2,888	+0.079	0.045	[−0.009, +0.168]	—
Wide pre (14–22, post 25+)	4,596	+0.028	0.035	[−0.040, +0.096]	—
Late post (15–20, post 28+)	1,777	+0.047	0.059	[−0.068, +0.162]	—
Much later (15–20, post 30+)	1,025	+0.417	0.225	[−0.024, +0.858]	—
UKHLS — has-party preference (0/1)					
Headline (pre 15–20, post 25+)	3,537	+0.057	0.025	[+0.008, +0.106]	**
Tight pre (16–19, post 25+)	3,080	+0.070	0.027	[+0.017, +0.124]	**
Wide pre (14–22, post 25+)	4,606	+0.034	0.022	[−0.008, +0.076]	—
Late post (15–20, post 28+)	1,856	+0.076	0.036	[+0.005, +0.146]	**
Much later (15–20, post 30+)	1,085	+0.081	0.041	[+0.000, +0.161]	**

Notes: Each row reports the Brand–Xie DRDID estimate of the tertiary ATT under a different pre/post window specification, computed under the 14-cov (SOEP) and 12-cov (UKHLS) headline PS. Headline = the specification used in the main text. "Tight pre" narrows the pre-window to ages 16–19; "wide pre" extends to ages 14–22. "Late post" and "much later" raise the post-window floor to ages 28+ and 30+ respectively. Significance: ** = 95% CI excludes zero. The headline ATT lies within the 95% CI of every alternative specification for the two significant SOEP outcomes and for the UKHLS has-party outcome.

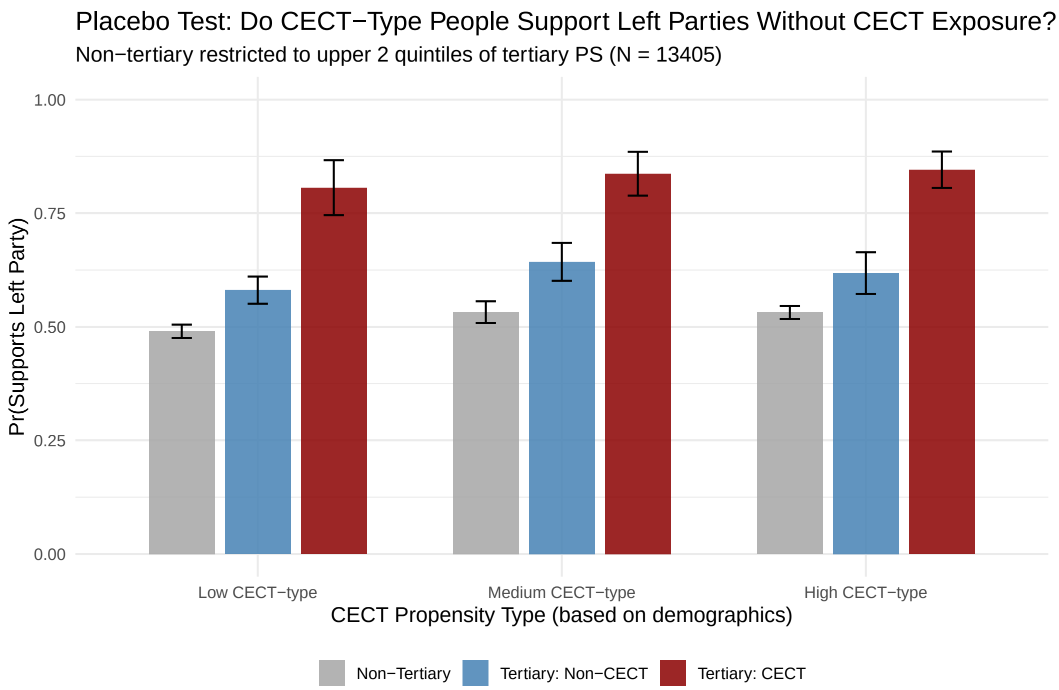

Figure 2: CECT pre-completion placebo: probability of holding a left-party preference, by CECT-propensity type and education group.

Notes: CECT-propensity type estimated from a field-choice propensity score (female, birth year, migrant background, Gymnasium track, religious attendance, parental education) among tertiary graduates and predicted onto the full sample. Three groups are compared at each CECT-propensity level: non-tertiary individuals (grey), tertiary graduates outside high-CECT fields (blue), and high-CECT tertiary graduates (red). The substantive contrast is between groups at the same CECT-propensity level: among individuals who would have been expected to choose high-CECT fields on demographics, do those who actually did so show a leftward shift relative to those who did not?

Appendix C. Supplementary results

This section reports supplementary tables and figures cross-referenced in the main text but not displayed in full. The Brand–Xie selection-on-observables reference layer (Appendix Tables 13 and 14) is the prior step that the local-IV, interactive fixed-effects, and Callaway–Sant'Anna estimators build on.

C.1 Full marginal-treatment-effect estimates by East/West subsample

The main text reports a slim two-outcome, two-subsample summary of the Zhou–Xie local-IV estimates. Table 19 reports the full $6 \times 3 \times 6$ grid: six outcomes (political interest, party identification, CDU/CSU, SPD/centre-left, Greens, Linke) $\times$ three subsamples (All Germany, West only, East only) $\times$ six estimands (ATE, TT, TUT, MPRTE, MPRTE-low, MPRTE-high). Bootstrap standard errors are from 500 person-level replicates. The East-only subsample is reported for completeness, but the small number of constituent East Bundesländer (six) yields confidence intervals too wide for independent inference. The all-Germany estimates for CDU/CSU and Linke are composition averages of opposite-sign within-subsample effects, as discussed in §.

C.2 Instrument-validity check: Bundesland fixed-effects DRDID on the Greens outcome

The local-IV identification of the propensity-varying Greens effect requires that, conditional on the covariates, birth Bundesland affects party support only through its effect on the probability of university attainment. The leading challenge to this exclusion restriction is the East/West divide: GDR socialisation generates cross-Bundesland baseline differences in party support that do not operate through HE attainment. Table 20 reports a Sant'Anna–Zhao doubly-robust DiD specification on the within-West sub-panel that adds Bundesland fixed effects to the outcome equation. The ATT falls from $+0.096$ (no FE) to $+0.074$ (with FE), a 23% reduction that leaves both the sign and the order of magnitude intact and rules out cross-Bundesland baseline differences as a

Table 19: Marginal-treatment-effect surface, by East/West subsample (local-IV)

Outcome	Subsample	ATE	Counterfactual avg. TT	Counterfactual avg. TUT	MPRTE at propensity median	MPRTE at propensity 25th	MPRTE at propensity 75th
Political interest							
	All Germany	$+1.04^{***}$	$+1.07^{***}$	$+1.00^{***}$	$+1.04^{***}$	$+1.09^{***}$	$+0.99^{***}$
		(0.16)	(0.17)	(0.20)	(0.16)	(0.17)	(0.17)
	West only	$+0.50^{**}$	$+0.39$	$+0.63^{**}$	$+0.50^{**}$	$+0.59^{**}$	$+0.43$
		(0.25)	(0.26)	(0.29)	(0.25)	(0.26)	(0.26)
	East only	$+0.80$	$+0.83^{*}$	$+0.78$	$+0.83$	$+0.87$	$+0.77$
		(0.50)	(0.47)	(0.58)	(0.51)	(0.56)	(0.51)
Has party preference							
	All Germany	$+0.53^{***}$	$+0.62^{***}$	$+0.43^{***}$	$+0.52^{***}$	$+0.55^{***}$	$+0.50^{***}$
		(0.10)	(0.11)	(0.12)	(0.11)	(0.11)	(0.11)
	West only	$+0.02$	$+0.15$	-0.13	-0.02	$+0.04$	-0.06
		(0.13)	(0.14)	(0.15)	(0.13)	(0.13)	(0.14)
	East only	$+0.68^{**}$	$+0.74^{***}$	$+0.63^{**}$	$+0.67^{**}$	$+0.64^{**}$	$+0.71^{**}$
		(0.27)	(0.26)	(0.30)	(0.27)	(0.29)	(0.28)
Leans CDU/CSU							
	All Germany	$+0.70^{***}$	$+0.75^{***}$	$+0.63^{***}$	$+0.68^{***}$	$+0.75^{***}$	$+0.63^{***}$
		(0.16)	(0.20)	(0.17)	(0.15)	(0.15)	(0.16)
	West only	$+0.38$	$+0.45$	$+0.25$	$+0.35$	$+0.35$	$+0.35$
		(0.29)	(0.33)	(0.28)	(0.28)	(0.28)	(0.29)
	East only	-0.34	-0.51	-0.14	-0.34	-0.22	-0.44
		(0.72)	(0.73)	(0.78)	(0.74)	(0.77)	(0.76)
Leans SPD (centre-left)							
	All Germany	-0.18	-0.15	-0.22	-0.23	-0.13	-0.28^{*}
		(0.15)	(0.18)	(0.16)	(0.14)	(0.15)	(0.15)
	West only	-0.77^{***}	-0.84^{***}	-0.63^{**}	-0.77^{***}	-0.72^{***}	-0.79^{***}
		(0.27)	(0.29)	(0.28)	(0.27)	(0.26)	(0.28)
	East only	$+0.34$	$+0.38$	$+0.30$	$+0.29$	$+0.43$	$+0.18$
		(0.45)	(0.44)	(0.52)	(0.47)	(0.50)	(0.47)
Leans Greens							
	All Germany	$+0.74^{***}$	$+0.63^{***}$	$+0.92^{***}$	$+0.85^{***}$	$+0.71^{***}$	$+0.94^{***}$
		(0.14)	(0.17)	(0.16)	(0.14)	(0.13)	(0.15)
	West only	$+0.56^{**}$	$+0.48^{**}$	$+0.71^{***}$	$+0.67^{***}$	$+0.54^{**}$	$+0.74^{***}$
		(0.22)	(0.23)	(0.25)	(0.23)	(0.21)	(0.25)
	East only	-0.02	-0.04	$+0.00$	$+0.02$	-0.01	$+0.05$
		(0.43)	(0.47)	(0.45)	(0.44)	(0.41)	(0.50)
Leans Linke							
	All Germany	-0.74^{***}	-0.64^{***}	-0.90^{***}	-0.82^{***}	-0.77^{***}	-0.85^{***}
		(0.12)	(0.13)	(0.13)	(0.12)	(0.12)	(0.13)
	West only	-0.29^{**}	-0.25^{*}	-0.36^{***}	-0.32^{**}	-0.31^{**}	-0.33^{**}
		(0.13)	(0.14)	(0.14)	(0.13)	(0.13)	(0.14)
	East only	$+0.31$	$+0.40$	$+0.20$	$+0.29$	$+0.16$	$+0.40$
		(0.38)	(0.39)	(0.45)	(0.39)	(0.44)	(0.40)

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$; bootstrap SEs (500 reps) in parentheses. Zhou–Xie local-IV MTE; instrument: birth Bundesland; controls: gender, birth year, migrant status, parental ISCED, parental ISEI, HE eligibility. Support trimmed to $[0.02, 0.98]$; estimates on the local-IV latent scale. East-only estimates ($n = 2{,}939$) are imprecise due to the small number of East Bundesländer; read as instrument-too-weak rather than precise nulls.

full explanation of the within-West Greens result.

Table 20: DRDID within-West Greens ATT, with and without Bundesland fixed effects

Specification	ATT	SE	p	95% CI	n
DRDID without Bundesland FE	+0.096	0.064	0.136	$[-0.030, +0.222]$	350
DRDID with Bundesland FE	+0.074	0.058	0.202	$[-0.039, +0.187]$	350

Estimator: Sant'Anna–Zhao doubly-robust DID (`DRDID::drdid`, panel=TRUE) on the two-period collapse of `supports_greens` using the pre-25 / post-25 age windows. Sample restricted to West-Germany birth Bundesländer.
Covariates in both specifications: gender, year of birth, migrant status, parental ISCED, parental ISEI (with missingness flag), HE-eligibility indicator. Specification with FE additionally enters `birthregion_f` (Bundesland of birth) as a factor in both the outcome and propensity-score models. `migrant` is collinear after the West-only restriction and is dropped automatically by `DRDID::drdid`.
After dropping sparse Bundesländer (cell $n < 50$ or zero treated/control), the analytic sample retains 4 West Bundesländer.

C.3 CS-DID robustness: alternative controls, cohort restrictions, and within-university field contrasts

Table 21 reports the staggered DiD ATT under six alternative specifications referenced in § as robustness: vocational-track controls (never-treated restricted to vocational-training completers); age ≤ 31 restriction; born ≥ 1980 cohort restriction; within-university CECT contrast (high-CECT vs low-CECT graduates, no never-treated comparison); within-university CECT under the alternative graduate-survey scoring of Van de Werfhost and De Graaf (45); and within-university CECT restricted to Hooghe-style completers only. The within-university CECT rows do not test the field-of-study channel of §, since the control group also receives the tertiary treatment; that test uses the Tertiary | High-CECT and Tertiary | Low-CECT specifications against non-attainers in Table 3.

Table 21: CS-DID robustness: alternative controls, cohort restrictions, and within-university field contrasts

Treatment	Outcome	ATT (SE)	N (T/C)	Robust at $\bar{M}$=0.5
Tertiary (vocational-track controls)				
	Political interest	$+0.058^{***}$ (0.020)	9,224 / 7,316	Yes
	Has party preference	$+0.055^{***}$ (0.011)	9,269 / 7,399	Yes
	Leans CDU/CSU	-0.029 (0.020)	5,781 / 3,369	Yes
	Leans SPD	$+0.045^{**}$ (0.018)	5,781 / 3,369	Yes
	Leans Greens	$+0.068^{***}$ (0.017)	5,781 / 3,369	Yes
	Leans Linke	-0.014 (0.011)	5,781 / 3,369	Yes
Tertiary (age $\leq$ 31 at observation)				
	Political interest	$+0.086^{***}$ (0.015)	6,835 / 21,998	Yes
	Has party preference	$+0.075^{***}$ (0.011)	6,878 / 24,292	Yes
	Leans CDU/CSU	-0.020 (0.015)	4,207 / 6,358	Yes
	Leans SPD	$+0.039^{***}$ (0.015)	4,207 / 6,358	Yes
	Leans Greens	$+0.019$ (0.013)	4,207 / 6,358	Yes
	Leans Linke	-0.005 (0.008)	4,207 / 6,358	Yes
Tertiary (born $\geq$ 1980 cohort)				
	Political interest	$+0.050^{***}$ (0.018)	6,962 / 24,241	Yes
	Has party preference	$+0.063^{***}$ (0.012)	6,988 / 28,211	Yes
	Leans CDU/CSU	-0.052^{***} (0.019)	4,296 / 6,888	Yes
	Leans SPD	$+0.051^{***}$ (0.020)	4,296 / 6,888	Yes
	Leans Greens	$+0.073^{***}$ (0.018)	4,296 / 6,888	Yes
	Leans Linke	-0.013 (0.012)	4,296 / 6,888	Yes
Within-university CECT contrast (high-CECT vs. low-CECT graduates)				
	Political interest	$+0.069^{**}$ (0.035)	1,322 / 2,752	Yes
	Has party preference	$+0.067^{***}$ (0.023)	1,328 / 2,810	Yes
	Leans CDU/CSU	-0.034 (0.039)	891 / 1,579	Yes
	Leans SPD	$+0.060$ (0.038)	891 / 1,579	Yes
	Leans Greens	$+0.011$ (0.032)	891 / 1,579	Yes
	Leans Linke	-0.011 (0.028)	891 / 1,579	Yes
Within-university CECT contrast (Werfhorst graduate-survey scoring)				
	Political interest	$+0.042$ (0.084)	528 / 1,716	Yes
	Has party preference	$+0.086^{*}$ (0.047)	528 / 1,740	Yes
	Leans CDU/CSU	-0.023 (0.047)	339 / 953	Yes
	Leans SPD	$+0.073$ (0.058)	339 / 953	Yes
	Leans Greens	$+0.010$ (0.056)	339 / 953	Yes
	Leans Linke	-0.096^{*} (0.053)	339 / 953	Yes
Within-university CECT contrast (completers only)				
	Political interest	$+0.138^{**}$ (0.060)	400 / 743	Yes
	Has party preference	$+0.092^{**}$ (0.041)	400 / 743	Yes
	Leans CDU/CSU	$+0.049$ (0.048)	322 / 468	Yes
	Leans SPD	-0.068^{*} (0.039)	322 / 468	Yes
	Leans Greens	$+0.066$ (0.049)	322 / 468	Yes
	Leans Linke	-0.048 (0.039)	322 / 468	Yes

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$; bootstrap SEs in parentheses. Callaway–Sant'Anna staggered DiD (IPW, universal base period). Final column reports whether the $\tau = +5$ ATT remains significant under Rambachan and Roth (32) relative-magnitude restrictions up to $\bar{M} = 0.5$.